MACHINE LEARNING FOUNDATIONS (MLF)

Comprehensive Categorized Past Year Question Repository (2022 – 2025)
Word-by-Word Verbatim Extraction & Official IIT Madras Syllabus Taxonomy

Total Unique Questions: 210	**Exam Sessions Covered:** 43 Papers	**Format:** MCQ, MSQ, NAT / SA
Week 1: ML Foundations (95 Qs)	**Week 2:** Math Foundations (22 Qs)	**Week 3:** Linear Algebra (63 Qs)
Week 4: Regression & Eigenvalues (30 Qs)	**Status:** Fully Deduplicated & Verified	**Interactive Mode:** Included (JEE CBT)

■ WEEK 1 - ML FOUNDATIONS

■ Category: Data & Models (2 Questions)

MLF-Q084 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

Which of the following is the best second-order polynomial that fits the dataset by the least sum of squares method? 3 ¯ (00 + OITi + min 00

Which of the following is the best second-order polynomial that fits the dataset by the least sum of squares method?

$$\min_{\theta_0,\theta_1,\theta_2} \sum_{i=1}^{3} (y_i - (\theta_0 + \theta_1 x_i + \theta_2 x_i^2))^2$$

x	1	0	-1
y	-1	1	1

(A) $2 - \frac{1}{2}x + \frac{5}{2}x^2$

(B) $2 - \frac{1}{2}x - \frac{5}{2}x^2$

(C) $2 + \frac{1}{2}x - \frac{5}{2}x^2$

(D) $2 + \frac{1}{2}x + \frac{5}{2}x^2$

Correct Answer: B

MLF-Q102 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

Let and be two lines in defined in parametric form: For example, every point on line is of the form . Which of the following is/are true?

$$L_1$$

$$L_2$$

and are parallel

(A) L_1

and are not parallel

(B) L_1

and intersect at a point

(C) L_1

and do not intersect at any point

(D) L_1

Correct Answer: B, D

■ Category: ML Concepts (80 Questions)

MLF-Q006 | [SA] | Marks: +5 | *Appeared in 1 papers*

Possible Answers : 4.56 to 4.76

The two eigenvalues of the matrix $\begin{bmatrix} 2 & 1 \\ 1 & p \end{bmatrix}$ have a ratio 3:1 for p=2. What is another value of p for which eigenvalues have the same ratio 3:1? (Enter the answer correct to two decimal places)

MLF-Q010 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the correct answer for a NOTE: Enter your answer to the nearest integer. Possible Answers : 1

Find the projection of vector $b = \begin{bmatrix} 2 \\ 3 \\ -1 \\ 2 \end{bmatrix}$ onto the line through vector $a = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix}$

If the answer is $\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$

MLF-Q011 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the correct answer for b NOTE: Enter your answer to the nearest integer. Possible Answers : 2

Find the projection of vector $b = \begin{bmatrix} 2 \\ 3 \\ -1 \\ 2 \end{bmatrix}$ onto the line through vector $a = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix}$

If the answer is $\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
8	$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
4	$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

MLF-Q012 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the correct answer for c NOTE: Enter your answer to the nearest integer. Possible Answers : -1

Find the projection of vector $b = \begin{bmatrix} 2 \\ 3 \\ -1 \\ 2 \end{bmatrix}$ onto the line through vector $a = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix}$

If the answer is $\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$

MLF-Q013 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the correct answer for d NOTE: Enter your answer to the nearest integer. Possible Answers : 3

Find the projection of vector $b = \begin{bmatrix} 2 \\ 3 \\ -1 \\ 2 \end{bmatrix}$ onto the line through vector $a = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix}$

If the answer is $\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$

MLF-Q014 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the value of a: (Enter answer correct to one decimal place) Possible Answers : 4

Find the second degree polynomial of the form $a + bx + cx^2$ which best fits the following data is

x	y
1	2
2	3
3	7

MLF-Q015 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the value of b: (Enter answer correct to one decimal place) Possible Answers : -3.6 to -3.4

Find the second degree polynomial of the form $a + bx + cx^2$ which best fits the following data is

x	y
1	2
2	3
3	7

MLF-Q016 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the value of c: (Enter answer correct to one decimal place) Possible Answers : 1.45 to 1.55

Find the second degree polynomial of the form $a + bx + cx^2$ which best fits the following data is

x	y
1	2
2	3
3	7

MLF-Q017 | [SA] | Marks: +2 | *Appeared in 1 papers*

Enter the value for m: NOTE: Enter your answer in one decimal place. Possible Answers : 2.8 to 3.0

The best fit line of the form mx+c using least squares method for the data set is

x	y
1	2
2	3
3	7
4	10
5	13

Note: Enter m and c of the best fit line respectively.

MLF-Q018 | [SA] | Marks: +2 | *Appeared in 1 papers*

Enter the value for c: NOTE: Enter your answer in one decimal place. Possible Answers : -1.8 to -1.6

The best fit line of the form mx+c using least squares method for the data set is

x	y
1	2
2	3
3	7
4	10
5	13

Note: Enter m and c of the best fit line respectively.

MLF-Q019 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the correct answer for a NOTE: Enter your answer to the nearest integer. Possible Answers : 6

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
8	$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
4	$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

MLF-Q020 | [SA] | Marks: +1 | *Appeared in 2 papers*

Enter the correct answer for c NOTE: Enter your answer to the nearest integer. Possible Answers : -2

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
8	$\begin{bmatrix}1\\1\end{bmatrix}$
4	$\begin{bmatrix}1\\-1\end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix}a & b\\c & d\end{bmatrix}$

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
-1	$\begin{bmatrix}1\\-1\end{bmatrix}$
-2	$\begin{bmatrix}1\\-2\end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix}a & b\\c & d\end{bmatrix}$

MLF-Q021 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the correct answer for d NOTE: Enter your answer to the nearest integer. Possible Answers : 6 MLT Number of Questions : 18 Section Marks : 50

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
8	$\begin{bmatrix}1\\1\end{bmatrix}$
4	$\begin{bmatrix}1\\-1\end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix}a & b\\c & d\end{bmatrix}$

MLF-Q028 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the value of a: NOTE: Enter your answer to the nearest integer. Possible Answers : 4

What is the linear approximation of $f(x_1, x_2) = 2x_1^2 + 2x_2^2$ around (1,1)?
If the approximation is $ax_1 + bx_2 + c$.

MLF-Q029 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the value of b: NOTE: Enter your answer to the nearest integer. Possible Answers : 4

What is the linear approximation of $f(x_1, x_2) = 2x_1^2 + 2x_2^2$ around (1,1)?
If the approximation is $ax_1 + bx_2 + c$.

MLF-Q030 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the value of c: NOTE: Enter your answer to the nearest integer. Possible Answers : -4

What is the linear approximation of $f(x_1, x_2) = 2x_1^2 + 2x_2^2$ around (1,1)?
If the approximation is $ax_1 + bx_2 + c$.

MLF-Q031 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the value of x: NOTE: Enter your answer in two decimal places. Possible Answers : 0.84 to 0.88

$$\text{Find the projection of vector } b = \begin{bmatrix} 2 \\ 3 \\ -1 \\ 2 \end{bmatrix} \text{ onto the line through vector } a = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}$$

NOTE: If the answer is $[x, y, z, w]^T$, then enter the values of x, y, z and w.

MLF-Q032 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the value of y: NOTE: Enter your answer in two decimal places. Possible Answers : 1.71 to 1.75

$$\text{Find the projection of vector } b = \begin{bmatrix} 2 \\ 3 \\ -1 \\ 2 \end{bmatrix} \text{ onto the line through vector } a = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}$$

NOTE: If the answer is $[x, y, z, w]^T$, then enter the values of x, y, z and w.

MLF-Q033 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the value of z: NOTE: Enter your answer in two decimal places. Possible Answers : 0.84 to 0.88

$$\text{Find the projection of vector } b = \begin{bmatrix} 2 \\ 3 \\ -1 \\ 2 \end{bmatrix} \text{ onto the line through vector } a = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}$$

NOTE: If the answer is $[x, y, z, w]^T$, then enter the values of x, y, z and w.

MLF-Q034 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the value of w: NOTE: Enter your answer in one decimal place. Possible Answers : 2.5 to 2.7

$$\text{Find the projection of vector } b = \begin{bmatrix} 2 \\ 3 \\ -1 \\ 2 \end{bmatrix} \text{ onto the line through vector } a = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}$$

NOTE: If the answer is $[x, y, z, w]^T$, then enter the values of x, y, z and w.

MLF-Q035 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the value of a: NOTE: Enter your answer to the nearest integer. Possible Answers : 5

The second degree polynomial of the form $a + bx + cx^2$ which best fits the following data is

x	y
1	3
2	4
3	8

MLF-Q036 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the value of b: NOTE: Enter your answer in one decimal place. Possible Answers : -3.6 to -3.4

The second degree polynomial of the form $a + bx + cx^2$ which best fits the following data is

x	y
1	3
2	4
3	8

MLF-Q037 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the value of c: NOTE: Enter your answer in two decimal places. Possible Answers : 1.45 to 1.55

The second degree polynomial of the form $a + bx + cx^2$ which best fits the following data is

x	y
1	3
2	4
3	8

MLF-Q038 | [SA] | Marks: +2 | *Appeared in 1 papers*

Enter the value of m: NOTE: Enter your answer in one decimal place. Possible Answers : 2.8 to 3.0

The best fit line of the form mx+c using least squares method for the data set is

x	y
1	3
2	4
3	8
4	11
5	14

Note: Enter m and c of the best fit line respectively.

MLF-Q039 | [SA] | Marks: +2 | *Appeared in 1 papers*

Enter the value of c: NOTE: Enter your answer in one decimal place. Possible Answers : -0.8 to -0.6

The best fit line of the form mx+c using least squares method for the data set is

x	y
1	3
2	4
3	8
4	11
5	14

Note: Enter m and c of the best fit line respectively.

MLF-Q040 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the correct answer for a NOTE: Enter your answer to the nearest integer. Possible Answers : 0

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
-1	$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
-2	$\begin{bmatrix} 1 \\ -2 \end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

MLF-Q041 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the correct answer for b NOTE: Enter your answer to the nearest integer. Possible Answers : 1

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
-1	$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
-2	$\begin{bmatrix} 1 \\ -2 \end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

MLF-Q042 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the correct answer for d NOTE: Enter your answer to the nearest integer. Possible Answers : -3 MLT Number of Questions : 18 Section Marks : 50

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
-1	$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
-2	$\begin{bmatrix} 1 \\ -2 \end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

MLF-Q043 | [SA] | Marks: +2 | *Appeared in 1 papers*

[Mathematical problem statement rendered in attached diagram below]

Correct Answer: 1.25

Explanation: *What the question asks We are told that the vector v We are given the matrix A eiaenvector of We must find the number a. I) is an Choose a question Answered C) Discussion*

MLF-Q044 | [SA] | Marks: +4 | *Appeared in 1 papers*

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 NOTE: Enter your answer to the nearest integer. Possible Answers : -21 Question Shuffling Allowed : Yes

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,4),(5,7)], i = 1$ to 5, Consider the regression model $f(x) = x + 2$. What is the mean squared loss of $f(x)$

For what value of k, will the following function be continuous?

$$f(x) = \begin{cases} 3x^2 - kx + 5, & \text{if } 0 \leq x < 2 \\ 1 - 3x, & \text{if } 2 \leq x \leq 3 \end{cases}$$

MLF-Q046 | [SA] | Marks: +4 | *Appeared in 1 papers*

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 NOTE: Enter your answer in one decimal place. Possible Answers : 3.3 to 3.7

What will be the directional derivative of $f(x, y, z) = x^2y^3 - 4xz$ at point (1, 1, 1) in the direction of [-1, 2, 0]?

MLF-Q047 | [SA] | Marks: +4 | *Appeared in 1 papers*

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 NOTE: Enter your answer in two decimal places. Possible Answers : 0.20 to 0.35

Find the linear approximation (L) of $f(x_1, x_2) = 2x_1^2 + 4x_2^4$ around (1, 1), and use it to compute L(1.1, 1.1). What will be the value of $f(1.1, 1.1) - L(1.1, 1.1)$?

MLF-Q049 | [SA] | Marks: +2 | *Appeared in 1 papers*

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Possible Answers : 0 Question Shuffling Allowed : Yes

Consider a function

$$f(x) = \begin{cases} \frac{x-|x|}{x}, & \text{if } x \neq 0 \\ 2, & \text{if } x = 0 \end{cases}$$

Is f continuous?

(Provide 1 as answer for 'Yes' and 0 for 'No'.)

MLF-Q050 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

Consider the system 2 5 7 Under what condition on bl, b2, b3 is this system solvable? Time : 0

$$\text{Consider the system } \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 2 & 4 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

Under what condition on b_1, b_2, b_3 is this system solvable?

b2 2b1

(A) $b_2 = 2b_1$

2 3bl

(B) $b_2 = 3b_1$

2bl

(C) $b_3 = 2b_1$

3bl

(D) $b_3 = 3b_1$

Correct Answer: C

MLF-Q053 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

Time : 0 Which of the following are the correct regarding the web server?

(A) A web server is a computer system capable of delivering web content over the

(B) A web server sends the response as an HTML document which is rendered on

(C) A web server listens for incoming network connections on a fixed port.

(D) A web server is a software that allows to send the request over the internet.

Correct Answer: A, B, C, D

MLF-Q065 | [SA] | Marks: +3 | *Appeared in 1 papers*

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Possible Answers : 2

For the data set $\{(x^i, y^i)\} = \{(1,1),(2,6),(3,8),(4,15),(5,26)\}, i = 1$ to 5, Consider the regression model $f(x) = x^2 + 1$. What is the mean squared loss of $f(x)$?(Round your answer off to the nearest integer).

MLF-Q066 | [SA] | Marks: +3 | *Appeared in 1 papers*

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Possible Answers : 0.2

What will be the average misclassification error when the functions $g(X) = sign((x_1 - x_2)^2 - 8)$ is used to classify the data points into classes $+1$ or -1. Enter the answer closest to one decimal accuracy.

X	y
(5,2)	1
(5,6)	-1
(6,3)	1
(6,4)	-1
(4,8)	-1

MLF-Q067 | [SA] | Marks: +3 | *Appeared in 1 papers*

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Possible Answers : 6 Question Shuffling Allowed : Yes Is Section Default? : null

For a 3×3 matrix A whose nullity is equal to 1, what is the value of $R(A) + \dim(C(A)) + \dim(C(A^T))$? Here $R(A)$ stands for Rank of A, $\dim(C(A))$ stands for dimension of column space of A, and $\dim(C(A^T))$ stands for dimension of column space of A^T. Enter the answer as an integer.

MLF-Q068 | [SA] | Marks: +2 | *Appeared in 1 papers*

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Possible Answers : 4 Java

If projection of vector A of length 10 units, onto vector B of length 8 units, has a magnitude of 5 units, then length of projection of vector B onto A is?

MLF-Q069 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

Time : 0 Match the URLs in column A with their correct rendered output in Column B with the correct app.py code for the flask application given in the previous question.

Column A	Column B
1. http://127.0.0.1:5000/greet?name=MADI&guest=MADII	a. **Hello, Guest!**
2. http://127.0.0.1:5000/greet?name=MADII&guest=MADI	b. **Hello, MADII!**
3. http://127.0.0.1:5000/greet?	c. **Hello, Guest!**
4. http://127.0.0.1:5000/greet?guest=MADI	d. **Hello, MADI!**

(A) 1-b, 2-d, 3-a, 4-c

(B) 1-b, 2-a, 3-c, 4-d

(C) 1-d, 2-b, 3-a, 4- c

(D) 1-d, 2-a, 3-c, 4-b

Correct Answer: C

MLF-Q073 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

Which of the following is the best approximation of s/fiGÄ? (Use linear approximation around 16). Time : 0

Which of the following is the best approximation of $\sqrt[4]{16.1}$? (Use linear approximation around 16).

(A) 2.1

(B) 2.003125

(C) 2.031

(D) 1.9968

Correct Answer: B

MLF-Q080 | [SA] | Marks: +4 | *Appeared in 3 papers*

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Possible Answers : 3.12 to 3.20 Java

Consider an encoder-decoder pair used for dimensionality reduction, where the encoder function is denoted as $f(x_1, x_2, x_3) = (x_1 + x_2 - x_3, x_1 + x_3)$ and the decoder function is denoted as $g(u) = (u_1, u_2, u_1 + u_2)$

Compute the reconstruction error $\left(R(f,g) = \frac{1}{n}\sum_{i=1}^{n} \|X_i - g(f(X_i))\|^2\right)$ for the following dataset:

$$X_1 = [1, 2, 3], \quad X_2 = [0, 2, 2], \quad X_3 = [1, 0, -1]$$

Enter the answer correct to two decimal places.

MLF-Q089 | [SA] | Marks: +2 | *Appeared in 3 papers*

Possible Answers : 15 Question Shuffling Allowed : Yes

Consider a matrix $A_{20\times25}$. If the nullity of the matrix is 10, then find the rank of the matrix.

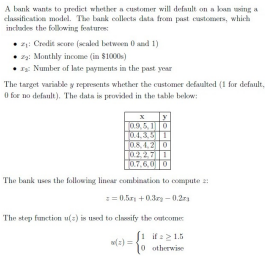
MLF-Q090 | [SA] | Marks: +3 | *Appeared in 3 papers*

Possible Answers : 6.37 to 6.43 Question Shuffling Allowed : No

If $f(x, y, z) = \sin(xy) + e^{yz} + xz^2$, what is the length of the gradient at the point, $(1, 0, 2)$? Enter the answer correct to two decimal places.

MLF-Q091 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

For the input x = [0.9, 5, 1], what will be the predicted output u(z)?



A bank wants to predict whether a customer will default on a loan using a classification model. The bank collects data from past customers, which includes the following features:

- x_1: Credit score (scaled between 0 and 1)
- x_2: Monthly income (in $1000s)
- x_3: Number of late payments in the past year

The target variable y represents whether the customer defaulted (1 for default, 0 for no default). The data is provided in the table below:

x	y
[0.9, 5, 1]	0
[0.4, 3, 3]	1
[0.8, 4, 2]	0
[0.2, 2, 7]	1
[0.7, 6, 0]	0

The bank uses the following linear combination to compute z:

$$z = 0.5x_1 + 0.3x_2 - 0.2x_3$$

The step function $u(z)$ is used to classify the outcome:

$$u(z) = \begin{cases} 1 & \text{if } z \geq 1.5 \\ 0 & \text{otherwise} \end{cases}$$

(A) 0

(B) 1

(C) 1.75

(D) Cannot be determined

Correct Answer: B, C

MLF-Q104 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

Suppose you are building a credit card fraud detection system and have two approaches: (i) Manually writing rules such as "If transaction amount is above Rs. 50,000 and the location is overseas, mark it as fraud." (ii) Training a model using millions of past transactions labeled as fraud or not fraud. Which of the following options is/are correct?

(A) Approach (i) is not machine learning.

(B) Approach (ii) is machine learning.

(C) Both approaches are machine learning.

(D) Neither approach is machine learning.

Correct Answer: A, B, C

MLF-Q107 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

Find . Your answer should be an integer. Possible Answers : 16

$$P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\mathbb{R}^2$$

MLF-Q108 | [MCQ] | Marks: +1 | *Appeared in 1 papers*

Find . Your answer should be an integer. Possible Answers : 4 Question Numbers : (16 to 17)

$$P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\mathbb{R}^2$$

MLF-Q119 | [SA] | Marks: +4 | *Appeared in 1 papers*

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Possible Answers : 5

Consider the following dataset with 5 data points:

X	y
$(2,3)$	5
$(-1,1)$	2
$(4,2)$	7
$(0,-2)$	1
$(-3,5)$	4

We want to fit a linear regression model of the form $f(X) = w^T x$ to this dataset, where $w = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Compute the value of the loss function L for this dataset which is defined as $L = \max_i |f(x^i) - y^i|$.

MLF-Q120 | [SA] | Marks: +4 | *Appeared in 1 papers*

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Possible Answers : 37 to 38

Let $f(x_1, x_2, x_3) = \frac{x_1 + x_2 + x_3}{2}$ be used as an encoder function and $g(u) = [u, u, u]$ be used as a decoder function for dimensionality reduction of the dataset $X_1 = [1, 2, 3]$, $X_2 = [2, 4, 6], X_3 = [3, 6, 9], X_4 = [4, 8, 12]$. Calculate the reconstruction error $\left(R(f, g) = \frac{1}{n}\sum_{i=1}^{n} \|x_i - g(f(x_i))\|^2\right)$ for this encoder decoder pair. Enter the answer correct to one decimal place.

MLF-Q122 | [SA] | Marks: +3 | *Appeared in 1 papers*

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Possible Answers : -1.16 to -1.12 Question Shuffling Allowed : Yes Is Section Default? : null

Find the directional derivative of $f(x, y, z) = x(x^2 - y^2) - z$ at a point D(1, -1, 0) in the direction of vector $\hat{h} = \frac{2}{7}i - \frac{3}{7}j + \frac{6}{7}k$. Enter the answer correct to two decimal places.

MLF-Q123 | [SA] | Marks: +2 | *Appeared in 1 papers*

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Possible Answers : 3.5 Question Shuffling Allowed : No Is Section Default? : null

Let the function $f(x) = \begin{cases} ax - 3, & x < 2 \\ 4, & x = 2 \\ 2x, & x > 2 \end{cases}$ be continuous for all $x \in \mathbb{R}$. Then, find the value of a.

Enter the answer correct to one decimal place.

MLF-Q124 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

Time : 0 Is the system Ax = b consistent?

$$\text{Let } A = \begin{pmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{pmatrix} \text{ and } b = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}.$$

(A) Yes

(B) No

Correct Answer: B

MLF-Q138 | [SA] | Marks: +3 | *Appeared in 3 papers*

Possible Answers : 1.6251 to 1.6258 Question Shuffling Allowed : Yes

Let $f(x, y) = \ln(x^2 + y^2)$. Use the linear approximation at the point $(1, 2)$ to estimate $f(1.02, 2.01)$. Write the answer correctly to 4 decimal places.

MLF-Q140 | [SA] | Marks: +3 | *Appeared in 4 papers*

Find the value of α + 2β. Possible Answers : 3

Let $A = \begin{pmatrix} 2 & 1 & 1 \\ a & 3 & 2 \\ 3 & b & c \end{pmatrix}$ be a 3×3 matrix. Let α be the eigenvalue corresponding to the eigenvector $v_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ and β is the eigenvalue corresponding to the eigenvector $v_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ of matrix A. Based on this information, answer the given sub-questions.

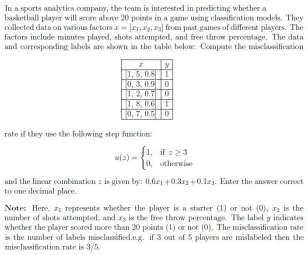
MLF-Q149 | [SA] | Marks: +3 | *Appeared in 4 papers*

Possible Answers : 20.58 to 20.64

If $f(x, y) = x^2y + 3xy^2$, what is the length of the gradient at the point (1,2)?
Enter the answer correct to two decimal places.

MLF-Q150 | [SA] | Marks: +3 | *Appeared in 4 papers*

Possible Answers : 0.2 Question Shuffling Allowed : Yes



In a sports analytics company, the team is interested in predicting whether a basketball player will score above 20 points in a game using classification models. They collected data on various factors $x = [x_1, x_2, x_3]$ from past games of different players. The factors include minutes played, shots attempted, and free throw percentage. The data and corresponding labels are shown in the table below: Compute the misclassification

x	y
[1, 5, 0.8]	1
[0, 3, 0.9]	0
[1, 2, 0.7]	0
[1, 8, 0.6]	1
[0, 7, 0.5]	0

rate if they use the following step function:

$$u(z) = \begin{cases} 1, & \text{if } z \geq 3 \\ 0, & \text{otherwise} \end{cases}$$

and the linear combination z is given by: $0.6x_1 + 0.3x_2 + 0.1x_3$. Enter the answer correct to one decimal place.

Note: Here, x_1 represents whether the player is a starter (1) or not (0), x_2 is the number of shots attempted, and x_3 is the free throw percentage. The label y indicates whether the player scored more than 20 points (1) or not (0). The misclassification rate is the number of labels misclassified.e.g. if 3 out of 5 players are mislabeled then the misclassification rate is 3/5.

MLF-Q152 | [MCQ] | Marks: +4 | *Appeared in 1 papers*

MSQ Correct Marks : 4 Max. Selectable Options : 0 Which of the following options is/are true ?

(A) [Option (A)]

(B) [Option (B)]

(C) [Option (C)]

(D) [Option (D)]

MLF-Q154 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

MSQ Correct Marks : 3 Max. Selectable Options : 0

(A) [Option (A)]

(B) [Option (B)]

(C) [Option (C)]

(D) [Option (D)]

MLF-Q155 | [MCQ] | Marks: +4 | *Appeared in 1 papers*

MSQ Correct Marks : 4 Max. Selectable Options : 0

(A) [Option (A)]

(B) [Option (B)]

(C) [Option (C)]

(D) [Option (D)]

MLF-Q156 | [MCQ] | Marks: +4 | *Appeared in 1 papers*

MCQ Correct Marks : 4

(A) [Option (A)]

(B) [Option (B)]

(C) [Option (C)]

(D) [Option (D)]

MLF-Q157 | [MCQ] | Marks: +4 | *Appeared in 1 papers*

MCQ Correct Marks : 4

(A) [Option (A)]

(B) [Option (B)]

(C) [Option (C)]

(D) [Option (D)]

MLF-Q164 | [SA] | Marks: +3 | *Appeared in 15 papers*

Possible Answers : 72

Consider the following dataset where each data point consists of three features x_1, x_2 and x_3 as

$$(2, 4, 3), (5, 5, 6), (2, 2, 1), (4, 4, 6)$$

Suppose the encoder and decoder functions, f and g respectively, are defined as

$$f(x_1, x_2, x_3) = 0.5x_1 + 0.5x_2 - x_3 \quad \text{and} \quad g(u) = [u, u, u].$$

Find the reconstruction loss.
Hint : "The reconstruction loss of the encoder decoder pair is the mean of the squared distance between the actual input and reconstructed input."

MLF-Q165 | [SA] | Marks: +3 | *Appeared in 15 papers*

Possible Answers : 1024 Question Shuffling Allowed : Yes

Does there exist a matrix A of shape 10×10 such that every non-zero vector in $\mathbb{R}^{10}$ is an eigenvector of A with eigenvalue 2? If yes, compute the value of $\det(A)$. If not, enter -1 as the answer.

MLF-Q166 | [SA] | Marks: +2 | *Appeared in 15 papers*

Possible Answers : 0 Question Shuffling Allowed : No

If the sum of the columns of a square matrix A is equal to the zero vector, then λ is certainly an eigenvalue of A. What is the value of λ?

Note that the term "column" here refers to a column vector, that is, each column of A is treated as a vector.

MLF-Q176 | [MCQ] | Marks: +2 | *Appeared in 15 papers*

Which of the following is

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Which of the following is $L_{(0,0)}[f](x)$?

(A) $1 + x_1 + x_2$

2.T'2

(B) $x_1 + 2x_2$

(C) $1 - x_1 - x_2$

(D) $x_1 - x_2$

Correct Answer: A

MLF-Q177 | [SA] | Marks: +2 | *Appeared in 15 papers*

Possible Answers : 1

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Recall that $L_{(0,0)}[f](x)$ corresponds to a plane in $\mathbb{R}^3$ and can be represented as the set of all points that satisfy the equation $w^T x + b = 0$, where $w \in \mathbb{R}^3, b \in \mathbb{R}$, with w being normal to the plane. If $w^T w = 3$, what is b?

MLF-Q178 | [MCQ] | Marks: +2 | *Appeared in 15 papers*

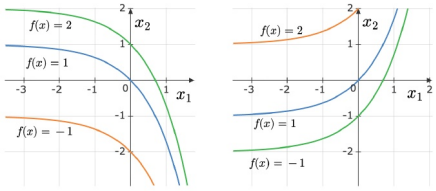
Which of the following represents the contours of f? (A) (B)

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Which of the following represents the contours of f?



(A) (B)

(A) (A)

(B) (B)

Correct Answer: A, B

MLF-Q179 | [SA] | Marks: +5 | *Appeared in 3 papers*

Possible Answers : 7 MLF

You are designing a sliding window protocol (assume Selective Repeat) with a data rate of 2 Mbps and a one-way delay of 0.5 seconds. Assuming each frame carries 2 kB of data, what is the minimum number of bits required for the sequence number?

MLF-Q180 | [MCQ] | Marks: +1 | *Appeared in 3 papers*

A bank wants to decide whether a loan applicant should be given a loan or not, based on features like income, credit score and past records. Which of the following machine learning techniques is applicable for this scenario?

MLF-Q185 | [SA] | Marks: +3 | *Appeared in 3 papers*

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Possible Answers : 0.25

Consider a regression model $h(x_1, x_2) = \dfrac{x_1 + x_2}{2}$. For the below input data points, find the mean squared loss of h. Enter the answer correct to two decimal places.

X	Y
[1, 2]	2
[2, 3]	3
[3, 4]	4
[4, 5]	5

The curve $y = ax^2 + bx + c$ passes through the points $(2, -1)$ and $(0, 0)$, and is tangent to the line $y = -x$ at the origin. Find the absolute value of the product ab. Enter the answer correct to two decimal places.

MLF-Q186 | [SA] | Marks: +3 | *Appeared in 3 papers*

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Possible Answers : 4.88 to 4.95

At the point $P_0 = (1, 2)$, the function $f(x, y)$ has a derivative of $2\sqrt{2}$ in the direction $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ and a derivative of 5 in the direction $(0, 1)$. Find the derivative of $f(x, y)$ at point P_0 in the direction $\left(\frac{-1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$. Enter the answer correct to two decimal places.

MLF-Q187 | [SA] | Marks: +3 | *Appeared in 3 papers*

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Possible Answers : 22 Question Shuffling Allowed : Yes Is Section Default? : null

Consider a 2×2 matrix A whose eigenvectors are $v_1 = [1, 1]^T$ and $v_2 = [1, 2]^T$ with eigenvalues 2 and 3, respectively. Find the sum of square of all the entries of A.

MLF-Q189 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

Find the second order approximation for f (x, y) = e2y¯r in the neighbourhood of (0, 0). Time : 0

Find the second order approximation for $f(x, y) = e^{2y-x}$ in the neighbourhood of $(0, 0)$.

(A) $1 - x + 2y - 2xy + y^2 + \frac{x^2}{2}$

2 1 - + 2y2 +

(B) $1 - 2xy + 2y^2 + \frac{x^2}{2}$

'2 2y2 +

(C) $2y^2 + \frac{x^2}{2}$

2

(D) $1 - x + 2y - 2xy + 2y^2 + \frac{x^2}{2}$

Correct Answer: D

MLF-Q195 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

[Mathematical problem statement rendered in attached diagram below]

MLF-Q196 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

[Mathematical problem statement rendered in attached diagram below]

MLF-Q197 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

[Mathematical problem statement rendered in attached diagram below]

MLF-Q203 | [MSQ] | Marks: +4 | *Appeared in 3 papers*

Determine which of the following limits yields a finite value.

(A) lim 1

$$\lim_{x\to 1} \frac{x^{\frac{1}{3}} - 1}{x^{\frac{1}{2}} - 1}$$

(B) lim

$$\lim_{x\to 1} \frac{1 - x^{\frac{1}{2}}}{1 - x}$$

(C) lim

$$\lim_{x\to \infty} \frac{x^2 - 7x}{x + 1}$$

(D) lim where a > 0

$$\lim_{x\to a} \frac{x^2 - a^2}{x^4 - a^4}, \quad \text{where } a > 0$$

(E) None of these

None of these

Correct Answer: A, B, D

MLF-Q208 | [SA] | Marks: +3 | *Appeared in 3 papers*

Possible Answers : 1.5

Consider the function, $f(x, y) = e^x \cos(y)$. Let $L(x, y)$ be the linear approximation of $f(x, y)$ at the point $(0, 0)$, then find the value $L(0.5, 0)$. Enter the answer correct to one decimal place.

MLF-Q209 | [SA] | Marks: +3 | *Appeared in 3 papers*

Possible Answers : 0.4 MLT

Let $f : \mathbb{R} \to \mathbb{R}$ be the function, $f(x) = 1 + e^{-x}$. What is the value of the derivative of f at x where $f(x) = 1.4$? Enter the answer correct to one decimal place.

■ Category: ML Tasks (8 Questions)

MLF-Q003 | [SA] | Marks: +4 | *Appeared in 2 papers*

The coach of Indian cricket team hired a ML team to predict whether Kohli will be successful or not in hitting a century in 2022 by using some classification models. Therefore the team collected presence/absence of various factors = [XI, an, xg] from both successful and unsuccessful innings by Kohli in past. The data and the corresponding label is shown in the table below. [0, 1,11 0 1 1 [1, 1, 11 1 Compute the zero-one error of the model if they have used and z = 0.5x1 — 0.8x2 + 0.4xg 1, 0, for z > 0 otherwise NOTE: Enter your answer to the nearest integer. Possible Answers : 0

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x	y
$[0, 1, 1]$	0
$[1, 0, 1]$	1
$[1, 0, 0]$	1
$[1, 1, 1]$	1

Compute the zero-one error of the model if they have used

$$u(z) = \begin{cases} 1, & \text{for } z \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

and $z = 0.5x_1 - 0.8x_2 + 0.4x_3$

The coach of indian cricket team hired a ML team to predict whether Kohli will be successful or not in hitting a century in 2022 by using some classification models. Therefore the team collected presence/absence of various factors $x = [x_1, x_2, x_3]$ from both successful and unsuccessful innings by Kohli in past. The data and the corresponding label is shown in the table below.

x	y
$[0, 1, 1]$	0
$[1, 0, 1]$	1
$[1, 0, 0]$	1
$[1, 1, 0]$	0

Compute the zero-one error of the model if they have used

$$u(z) = \begin{cases} 1, & \text{for } z \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

and $z = 0.5x_1 - 0.8x_2 + 0.4x_3$

MLF-Q004 | [SA] | Marks: +5 | *Appeared in 2 papers*

Consider this dimensionality reduction problem, the encoder and decoder = [u, —u, u] respectively are used functions expressed as f(:r) = -+- — and g(u) to reduce the dimension of the data set given below 1] [1, 2, 11 What is the mean resconstruction error obtained for the model? NOTE: Enter your answer to the nearest integer. Possible Answers : 18 to 20

Consider this dimensionality reduction problem, the encoder and decoder functions expressed as $f(x) = x_1 + x_2 - x_3$ and $g(u) = [u, -u, u]$ respectively are used to reduce the dimension of the data set given below

x
$[-1, 1, -1]$
$[1, 2, 1]$
$[1, 1, -1]$
$[0, 1, 3]$

What is the mean resconstruction error obtained for the model?

Consider this dimensionality reduction problem, the encoder and decoder functions expressed as $f(x) = x_1 + x_2 - x_3$ and $g(u) = [u, -u, u]$ respectively are used to reduce the dimension of the data set given below

x
$[-1, 1, -1]$
$[1, 2, 1]$
$[1, 1, -1]$
$[0, 1, 2]$

What is the mean resconstruction error obtained for the model?

MLF-Q048 | [MSQ] | Marks: +2 | *Appeared in 1 papers*

Time : 0 Identify which of the following requires use of classification technique.

(A) Predicting the amount of rainfall in May 2022 in North India based on

(B) Predicting the price of a land based on its area and distance from the market.

(C) Predicting whether an email is spam or not.

(D) Predicting whether the patient is having cancer or not.

Correct Answer: C, D

MLF-Q055 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

Time : 0 Which of the following statements is/are true?

(A) Email spam detection falls under classification problem.

(B) Predicting the price of a house based on previous datasets falls under

(C) Classifying whether handwriting belongs to a particular person or not based

(D) Finding out the probability that a particular poem is written by Shakespeare

Correct Answer: A, B, C, D

MLF-Q139 | [MCQ] | Marks: +2 | *Appeared in 3 papers*

A hospital collects patient data including symptoms, vital signs, and lab test results. They want to find patterns to improve treatment plans. What type of task is clustering most suitable for?

(A) Predicting the recovery time for a new patient.

(B) Diagnosing diseases based on labeled medical records.

(C) Grouping patients with similar symptoms and test results without prior labels.

(D) Determining the most important variable for predicting disease.

Correct Answer: C, D

MLF-Q151 | [MCQ] | Marks: +2 | *Appeared in 3 papers*

Consider a dataset from an e-commerce company such as Flipkart that has details of customer behavior, which of the following problems can be best approached using clustering?

(A) Predicting the exact number of sales for each customer segment.

(B) Classifying customers into segments based on labeled customer data.

(C) Identifying inherent groupings in customer data without predefined labels.

(D) Reducing the number of features in the customer dataset.

Correct Answer: C, D

MLF-Q160 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

MCQ Correct Marks : 2 Consider a dataset from an e-commerce company such as Flipkart that has details of customer behavior, which of the following problems can be best approached using clustering? Options : 6406532782149. Predicting the exact number of sales for each customer segment. 6406532782150. Classifying customers into segments based on labeled customer data. 6406532782151. Identifying inherent groupings in customer data without predefined labels. 6406532782152. Reducing the number of features in the customer dataset. Business Analytics

MLF-Q161 | [MCQ] | Marks: +1 | *Appeared in 15 papers*

A bank wants to decide whether a loan applicant should be given a loan or not, based on features like income, credit score and past records. Which of the following machine learning techniques is applicable for this scenario?

(A) Regression

(B) Classification

(C) Clustering

(D) Dimensionality Reduction

Correct Answer: B, C

■ Category: Types of Learning (2 Questions)

MLF-Q076 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

Time : 0 Which of the following is/are supervised learning task(s)?

(A) Predicting future sales of a product based on the past sales data.

(B) Reducing the dimensionality of images while retaining essential features,

(C) Recognizing and interpreting handwritten characters or words, often used in

(D) Categorizing text documents into predefined classes, such as spam or non-

Correct Answer: A, C, D

MLF-Q181 | [MSQ] | Marks: +2 | *Appeared in 3 papers*

Time : 0 Which of the following is/are supervised learning task(s)?

(A) Predicting future weather conditions based on the past weather observations.

(B) Reducing the dimensionality of images while retaining essential features,

(C) Identifying objects in images with multiple categories, such as cats, dogs, cars,

(D) Categorizing news articles into topics like politics, sports, entertainment, and

Correct Answer: A, C, D

■ WEEK 2 - MATHEMATICAL FOUNDATIONS

■ Category: Continuity (3 Questions)

MLF-Q005 | [SA] | Marks: +5 | *Appeared in 1 papers*

Find the value of a such that f (x) is continuous at a: if — 1 < < 0 a: if 0 < x < 1 NOTE: Enter your answer to the nearest integer. Possible Answers : -1

$$\text{Find the value of } a \text{ such that } f(x) \text{ is continuous at } x = 0:$$

$$f(x) = \begin{cases} \dfrac{\sqrt{1+ax} - \sqrt{1-ax}}{x}, & \text{if } -1 \le x < 0 \\ \dfrac{2x+1}{x-1}, & \text{if } 0 \le x < 1 \end{cases}$$

MLF-Q024 | [SA] | Marks: +5 | *Appeared in 1 papers*

Find the value of a such that is continuous at x 4.1B 2 if 0 < < 2 NOTE: Enter your answer to the nearest integer. Possible Answers : -4

Find the value of a such that $f(x)$ is continuous at $x = 0$:

$$f(x) = \begin{cases} \dfrac{\sqrt{4+ax} - \sqrt{4-ax}}{x}, & \text{if } -2 \le x < 0 \\ \dfrac{4x+2}{4x-1}, & \text{if } 0 \le x < 2 \end{cases}$$

MLF-Q206 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

For what value of b is :r—b continuous at every x? if < 0 if > 0

For what value of b is

$$f(x) = \begin{cases} \dfrac{x-b}{b+1}, & \text{if } x < 0 \\ x^2 + b, & \text{if } x \ge 0 \end{cases}$$

continuous at every x?

(A) 3

(B) 2

(C) -2

(D) 1

Correct Answer: C, D

■ Category: Derivatives (15 Questions)

MLF-Q026 | [SA] | Marks: +3 | *Appeared in 1 papers*

Let f (x, y, z) = —3r + 4ye6z. Find directional derivative of fat (1,-1,0) in the direction of unit vector along (1,0,-1) NOTE: Enter your answer in one decimal place. Possible Answers : 14.5 to 15.0

Let $f(x, y, z) = -3x + 4ye^{6z}$. Find directional derivative of f at (1,-1,0) in the direction of unit vector along (1,0,-1)

MLF-Q056 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

Which of the following statements is/are true about function f (x) = Time : 0

Which of the following statements is/are true about function $f(x) = \begin{cases} \frac{|x|}{x^2}, x \neq 0 \\ 0, x = 0 \end{cases}$?

(A) The function is differentiable for all the values of x except when x = 0.

(B) The function is differentiable for all the values of x.

(C) The function is continuous at x = 0.

(D) The function is not continuous at x = 0.

Correct Answer: A, B, D

MLF-Q075 | [MCQ] | Marks: +4 | *Appeared in 3 papers*

Let f : R2 —+ R be a multivariable scalar valued continuous function, such that Df 2x + 1 and Suppose lim Find out the directional derivative of f at the point (0, 0) in the direction of the vector (1, 1). Time : 0

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a multivariable scalar-valued continuous function, such that

$\frac{\partial f}{\partial x} = 2x + 1$ and $\frac{\partial f}{\partial y} = 2y + 1$.

Suppose $\lim_{(x,y)\to(0,0)} f(x, y) = 5$.

Find out the directional derivative of f at the point $(0, 0)$ in the direction of the vector $(1, 1)$.

(A) $\sqrt{2}$

(B) $2\sqrt{2}$

(C) $-2\sqrt{2}$

(D) $-\sqrt{2}$

Correct Answer: A

MLF-Q079 | [MSQ] | Marks: +4 | *Appeared in 3 papers*

Consider a function f : R —+ R defined as f (a:) = lx(x — Which of the following options is/are true? Time : 0

Consider a function $f : \mathbb{R} \longrightarrow \mathbb{R}$ defined as $f(x) = |x(x-2)|$
Which of the following options is/are true?

(A) lim f(:r) = lim f(:r).
$\lim_{x \to 2^+} f(x) = \lim_{x \to 2^-} f(x).$

(B) f is differentiable at a; = 0.
f is differentiable at $x = 0$.

(C) f'(3) = 4.
$f'(3) = 4.$

(D) f is continuous at a: = 0.
f is continuous at $x = 0$.

Correct Answer: A, C, D

MLF-Q083 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

The directional derivative of a scalar function f (x, y) at a point (To, yo) in the direction of a unit vector u represents:

The directional derivative of a scalar function $f(x, y)$ at a point (x_0, y_0) in the direction of a unit vector u represents:

(A) [Option (A)]

(B) The magnitude of the gradient vector Vf at (To, yo).
The magnitude of the gradient vector ∇f at (x_0, y_0).

(C) The maximum value of the gradient at (To, yo).
The maximum value of the gradient at (x_0, y_0).

(D) The angle between the vector u and V f.
The angle between the vector u and ∇f.

Correct Answer: A

MLF-Q087 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

Which of the following statements is/are correct?

(A) f (x) = In(:r) is continuous and differentiable in the domain 0,
$f(x) = \ln(x)$ is continuous and differentiable in the domain $[0, \infty)$.

(B) f(:r) e is continuous and differentiable for all e R.
$f(x) = e^x$ is continuous and differentiable for all $x \in \mathbb{R}$.

(C) f(:r) = is continuous but not differentiable at x = 0.
$f(x) = \sqrt{x}$ is continuous but not differentiable at $x = 0$.

(D) f ( ) — is continuous and differentiable for all 0.
$f(x) = \frac{1}{x}$ is continuous and differentiable for all $x \neq 0$.

Correct Answer: B, C, D

MLF-Q096 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

Consider the function : Which of the following is true?

$$f : \mathbb{R} \rightarrow \mathbb{R}$$

$$f(x) = \begin{cases} x^2 + 2, & x \leqslant 1 \\ x + 2, & x > 1 \end{cases}$$

is continuous at , but not differentiable at .

(A) $x = 1$

is both continuous and differentiable at .

(B) $x = 1$

is neither continuous nor differentiable at .

(C) $x = 1$

is differentiable at , but not continuous at .

(D) $x = 1$

Correct Answer: A

MLF-Q106 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

Find the directional derivative of the function given by at the point in the direction of the vector . Your answer should be an integer. Possible Answers : 3 Question Shuffling Allowed : No Question Numbers : (14 to 15)

$$f : \mathbb{R}^3 \rightarrow \mathbb{R}$$

$$f(x, y, z) = x^2 y - y z^3 + z$$

MLF-Q118 | [MCQ] | Marks: +4 | *Appeared in 1 papers*

Let f : 122 R, given by y) = -Fy2. Consider the following algorithm: xn+l = Xn - hVf(Xn), : 1,2,.. and h > 0, where h is the step size and Vf(Xn) is the gradient of f evaluated at Xn. Fix h = 0.25 and Xo = (1, 3), which among the following points does the algorithm converge to? Time : 0

Let $f : \mathbb{R}^2 \to \mathbb{R}$, given by $f(x, y) = x^2 + y^2$. Consider the following algorithm:

$$X_{n+1} = X_n - h\nabla f(X_n), \quad X_i \in \mathbb{R}^2, i : 1, 2, \ldots \text{ and } h > 0,$$

where h is the step size and $\nabla f(X_n)$ is the gradient of f evaluated at X_n. Fix $h = 0.25$ and $X_0 = (1, 3)$, which among the following points does the algorithm converge to?

(A) (1, 3)

(B) (1/2, 1/6)

(C) (0, 0)

(D) The algorithm does not converge

Correct Answer: C

MLF-Q130 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

The function is differentiable at x must be true? px2 4- +2, if x < 1 = 1. If p, q, r, and s are constants, which of the following is/are

The function

$$f(x) = \begin{cases} px^2 + qx + 2, & \text{if } x < 1 \\ rx + s, & \text{if } x \geq 1 \end{cases}$$

is differentiable at $x = 1$. If p, q, r, and s are constants, which of the following is/are must be true?

(A) r = 2p + q and s = 2 — p

$r = 2p + q$ and $s = 2 - p$

(B) $p + q = r + s$

(C) q = r and s — -2

$q = r$ and $s = 2$

(D) p + q 2 = r + s 21) q = r

$p + q + 2 = r + s$ and $2p + q = r$

Correct Answer: A, D

MLF-Q146 | [MCQ] | Marks: +4 | *Appeared in 3 papers*

The function be true? ax2 + b:r+l, if < 0 if x > 0 is differentiable at (x — 0). If a, b, c, and d are constants, which of the following must

The function

$$f(x) = \begin{cases} ax^2 + bx + 1, & \text{if } x < 0 \\ cx + d, & \text{if } x \geq 0 \end{cases}$$

is differentiable at $(x = 0)$. If a, b, c, and d are constants, which of the following must be true?

a = c and b

(A) $a = c \text{ and } b = 1$

b = c and d — —1

(B) $b = c \text{ and } d = 1$

a = c and d

(C) $a = c \text{ and } d = 1$

(D) $b = d = 1$

Correct Answer: B

MLF-Q147 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

Which of the following statements is true for a function f(x, y) that is differentiable at a point (a, b)?

(A) The directional derivative exists in every direction.

(B) The directional derivative is the same in every direction.

(C) The directional derivative can only be calculated along the axes.

(D) The directional derivative does not exist.

Correct Answer: A

MLF-Q158 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

MCQ Correct Marks : 3 Which of the following statements is true for a function f(x, y) that is differentiable at a point (a, b)?

(A) The directional derivative exists in every direction.

(B) The directional derivative is the same in every direction.

(C) The directional derivative can only be calculated along the axes.

(D) The directional derivative does not exist.

MLF-Q188 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

— l/x2 Let f(:r) o, :r>() Select which among the following are true? Time : 0

$$\text{Let } f(x) = \begin{cases} e^{-1/x^2}, & x > 0 \\ 0, & x \leq 0 \end{cases}. \text{ Select which among the following are true?}$$

(A) f is discontinuous at x = 0.

(B) f is differentiable.

(C) f is not differentiable at x = 0.

(D) f'(0) = 0

Correct Answer: B, D

MLF-Q207 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

Consider the function, f (x, y) = -+- .3y2, in which direction does the maximum directional derivative occur at the point (1, 2)?

Consider the function, $f(x, y) = x^2 + 3y^2$, in which direction does the maximum directional derivative occur at the point $(1, 2)$?

(A) $\left(\frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}}\right)$

(B) $\left(\frac{1}{\sqrt{37}}, \frac{6}{\sqrt{37}}\right)$

2 24)

(C) $\left(\frac{2}{\sqrt{5}}, \frac{24}{\sqrt{5}}\right)$

(D) $\left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$

Correct Answer: B

■ Category: Functions (12 Questions)

MLF-Q008 | [SA] | Marks: +1 | *Appeared in 1 papers*

Compute the equation of the function's linear approximation at point (2,3). If the equation of the linear approximation is f(x, y) = ax + by + c, then provide the values of b. NOTE: Enter your answer in one decimal place. Possible Answers : -0.8 to -0.7

$$\text{Consider the function } f(x, y) = \sqrt{41 - 4x^2 - y^2}.$$

MLF-Q009 | [SA] | Marks: +3 | *Appeared in 1 papers*

Give the maximum rate of change of the function at (2,3). NOTE: Enter your answer in one decimal place. Possible Answers : 2.0 to 2.5

$$\text{Consider the function } f(x, y) = \sqrt{41 - 4x^2 - y^2}.$$

MLF-Q054 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

Time : 0 Which of the following is/are true about loss after applying Encoder and Decoder functions to a given data?

(A) The loss value can be equal to any real number.

(B) The loss value cannot be negative.

(C) A perfect encoder and decoder function will not exist for any data, that is loss

(D) For some data, there can exist a perfect encoder and decoder functions, that is

Correct Answer: B, C, D

MLF-Q059 | [MCQ] | Marks: +4 | *Appeared in 1 papers*

A quadratic approximation of the function f = sm :r at = — radian is: 4 Time : 0

$$\text{A quadratic approximation of the function } f(x) = \sin^2 x \text{ at } x = \frac{\pi}{4} \text{ radian is:}$$

4

(A) $Q_{\frac{\pi}{4}}[f(x)] = \frac{1}{2} + (x - \frac{\pi}{4}) + (x - \frac{\pi}{4})$

(B) $Q_{\frac{\pi}{4}}[f(x)] = \frac{1}{2} + (x - \frac{\pi}{4})$

(C) $Q_{\frac{\pi}{4}}[f(x)] = \frac{1}{2} + 2(x - \frac{\pi}{4}) + \frac{1}{2}(x - \frac{\pi}{4})$

(D) $Q_{\frac{\pi}{4}}[f(x)] = \frac{\pi}{2} + 2(x - \frac{\pi}{4}) + \frac{1}{2}(x - \frac{\pi}{4})$

Correct Answer: B

MLF-Q060 | [MCQ] | Marks: +4 | *Appeared in 1 papers*

The direction of steepest descent for the function y, z) at the point (r, y, z) = = cos(x) sin(y) cos(z) Time : 0

$$\text{The direction of steepest descent for the function } f(x, y, z) = \cos(x)\sin(y)\cos(z)$$
$$\text{at the point } (x, y, z) = \left(\frac{\pi}{4}, \frac{\pi}{6}, \frac{\pi}{4}\right) \text{ is?}$$

(A) $\left[-\frac{1}{4}, \frac{\sqrt{3}}{4}, -\frac{1}{4}\right]$

(B) $\left[\frac{1}{4}, -\frac{\sqrt{3}}{4}, \frac{1}{4}\right]$

(C) $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$

(D) $\left[-\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}\right]$

Correct Answer: B

MLF-Q061 | [MCQ] | Marks: +4 | *Appeared in 1 papers*

Which of the following is the equation of the line that passes through point [1, 2, 3, 4] and is tangent to the function f(p, q, r, s) = p2+q -Fr2 + s2 -Jrpq -Frs -2p-2q at the point [p q r s] Time : 0

Which of the following is the equation of the line that passes through point $[1, 2, 3, 4]$ and is tangent to the function $f(p, q, r, s) = p^2+q^2+r^2+s^2+pq+rs-2p-2q$ at the point $[p, q, r, s] = [1, 3, 5, 1]$?

(A) [1, + 5, 11, 7]

$[1, 2, 3, 4] + \alpha[3, 5, 11, 7]$

(B) 13, 5,

$[3, 5, 11, 7] + \alpha[1, 2, 3, 4]$

(C) [1, 2, 3, 4] + 4, 10, 2]

$[1, 2, 3, 4] + \alpha[0, 4, 10, 2]$

(D) [1, 2, 3, 4] + 3, 11, 7]

$[1, 2, 3, 4] + \alpha[5, 3, 11, 7]$

(E) $[5, 3, 11, 7] + \alpha[1, 2, 3, 4]$

Correct Answer: A

MLF-Q081 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

= sin(xy). Let y) be the linear approximation Consider the function f (x, y) of f (x, y) around the point (9, 0). What is the value of L + 0.1, 0.1)? hint: Use the approximation of = 3.14

Consider the function $f(x, y) = \sin(xy)$. Let $L(x, y)$ be the linear approximation of $f(x, y)$ around the point $\left(\frac{\pi}{2}, 0\right)$. What is the value of $L\left(\frac{\pi}{2}+0.1, 0.1\right)$?

hint: Use the approximation of $\pi = 3.14$

(A) 0.162

(B) 0.257

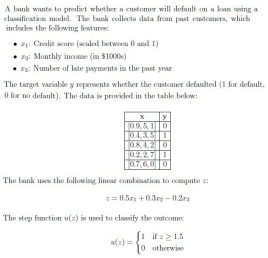
(C) 0.157

(D) 1.57

Correct Answer: C, D

MLF-Q092 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

If the threshold in the step function is changed to z ≥ 1.0, what will happen to the misclassification rate?



A bank wants to predict whether a customer will default on a loan using a classification model. The bank collects data from past customers, which includes the following features:

- x_1: Credit score (scaled between 0 and 1)
- x_2: Monthly income (in $1000s)
- x_3: Number of late payments in the past year

The target variable y represents whether the customer defaulted (1 for default, 0 for no default). The data is provided in the table below:

x	y
[0.9, 5, 1]	0
[0.4, 3, 5]	1
[0.8, 4, 2]	0
[0.2, 2, 7]	1
[0.7, 6, 0]	0

The bank uses the following linear combination to compute z:

$$z = 0.5x_1 + 0.3x_2 - 0.2x_3$$

The step function $u(z)$ is used to classify the outcome:

$$u(z) = \begin{cases} 1 & \text{if } z \geq 1.5 \\ 0 & \text{otherwise} \end{cases}$$

(A) Increase

(B) Decrease

(C) Remain same

(D) Cannot be determined

Correct Answer: A, B

MLF-Q105 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

Consider the function given by . If the best linear approximation to at is given as , find . Your answer should be an integer. Possible Answers : 11 Question Shuffling Allowed : Yes

$$f(x) = \frac{1}{(1+2x)^5}$$

$$f : \mathbb{R} \setminus \left\{ \frac{-1}{2} \right\} \rightarrow \mathbb{R}$$

MLF-Q117 | [MCQ] | Marks: +4 | *Appeared in 1 papers*

Find the linear approximation of the function f (x, y) neighbourhood of (0, 1). = yer Yin the Time : 0

Find the linear approximation of the function $f(x,y) = ye^{xy}$ in the neighbourhood of (0,1).

(A) $x + y$

(B) $x + 2y$

(C) $2x + y$

(D) $2x + 3y$

Correct Answer: A

MLF-Q148 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

Which of the following functions cannot be approximated linearly around x = 0?

f(x) = sin@)

(A) $f(x) = \sin(x)$

= cos(:r)

(B) $f(x) = \cos(x)$

f (x) = In(x)

(C) $f(x) = \ln(x)$

2

(D) $f(x) = x^2$

Correct Answer: C

MLF-Q159 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

MCQ Correct Marks : 3 Which of the following functions cannot be approximated linearly around x = 0?

(A) [Option (A)]

(B) [Option (B)]

(C) [Option (C)]

(D) [Option (D)]

■ Category: Sequences & Limits (1 Questions)

MLF-Q137 | [MSQ] | Marks: +2 | *Appeared in 3 papers*

Which of the following is/are true?

(A) If a = —I then an O as n 00.
If $a_n = -1 + \frac{n}{n+1}$ then $a_n \to 0$ as $n \to \infty$.

(B) If an = Sin (n) then an O as n 00.
If $a_n = \sin(n)$ then $a_n \to 0$ as $n \to \infty$.

(C) If a = n! then an 00 as n 00.
If $a_n = n!$ then $a_n \to \infty$ as $n \to \infty$.

(D) an = sin (n) is not a convergent sequence.
$a_n = \sin(n)$ is not a convergent sequence.

Correct Answer: A, C, D

■ Category: Sets & Number Systems (2 Questions)

MLF-Q001 | [SA] | Marks: +3 | *Appeared in 1 papers*

The decimal representation of the octal number 7128 is NOTE: Enter your answer to the nearest integer. Possible Answers : 458 MLF Number of Questions : 12 Section Marks : 50

The decimal representation of the octal number 712_8 is ___.

MLF-Q022 | [SA] | Marks: +3 | *Appeared in 1 papers*

The decimal representation of the octal number 7448 is NOTE: Enter your answer to the nearest integer. Possible Answers : 484 MLF Number of Questions : 13 Section Marks : 50

The decimal representation of the octal number 744_8 is ___.

■ WEEK 3 - LINEAR ALGEBRA & SUBSPACES

■ Category: Dot Product & Norms (6 Questions)

MLF-Q095 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

Let the vectors be orthogonal. Which of the following must be true?

$$u = (1, a, 2), v = (2, -1, b)$$

(A) $a - 2b = 2$

(B) $a + 2 = 2b$

2b

(C) $2b - a = 2$

(D) $a - 2b = 0$

Correct Answer: A

MLF-Q127 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

Which of the following options is/are true regarding the projection matrix?

(A) It is a symmetric matrix.

(B) It is an orthogonal matrix.

(C) It is orthogonally diagonalizable.

(D) It is always a rank 1 matrix.

Correct Answer: A, B, C, D

MLF-Q134 | [MSQ] | Marks: +2 | *Appeared in 3 papers*

Which of the following options is/are true?

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

(A) A* = A, where A* is conjugate transpose of matrix A.

(B) A is a symmetric matrix.

(C) A is a unitary matrix.

(D) Columns of A are not orthogonal.

Correct Answer: B, C, D

MLF-Q163 | [MSQ] | Marks: +2 | *Appeared in 15 papers*

(1, 2, and Two vectors VI and v2 are such that VI Which of the following options is/are correct?

Two vectors v_1 and v_2 are such that $v_1 = (1, 2, 2)^T$ and $v_2 = (2, -1, 0)^T$.
Which of the following options is/are correct?

(A) Vectors VI and c'2 are parallel to each other.
Vectors v_1 and v_2 are parallel to each other.

(B) Vectors VI and v2 are orthogonal to each other.
Vectors v_1 and v_2 are orthogonal to each other.

(C) Projection of VI onto t'2 is the zero vector.
Projection of v_1 onto v_2 is the zero vector.

(D) Projection of t'2 onto VI is the zero vector.
Projection of v_2 onto v_1 is the zero vector.

Correct Answer: B, C, D

MLF-Q171 | [MSQ] | Marks: +2 | *Appeared in 15 papers*

Which of the following is/are true?

Consider the matrix A given below:

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

where $0 < \theta < 180°$.

(A) A is an orthogonal matrix

(B) A is invertible

(C) A is symmetric

Correct Answer: A, B, C

MLF-Q201 | [MSQ] | Marks: +4 | *Appeared in 3 papers*

Consider two vectors a = 1 —1 and b = 0 1 2 Let P be the projection —1 matrix that projects a vector onto the line through the vector a. Which of the following options is/are true?

Consider two vectors $a = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Let P be the projection matrix that projects a vector onto the line through the vector a. Which of the following options is/are true?

(A) P is symmetric matrix.

P is symmetric matrix.

(B) Projection of b onto a is —

Projection of b onto a is $\frac{1}{4}(-a)$.

(C) P is rank one matrix.

P is rank one matrix.

(D) Plane = y is orthogonal to a.

Plane $x = y$ is orthogonal to a.

Correct Answer: A, C, D

Category: Matrices (15 Questions)

MLF-Q007 | [SA] | Marks: +3 | *Appeared in 1 papers*

Find the element 122.2 of the projection matrix P that project on to the line —1 7 . (Assume matrix entries are indexed starting from (1, 1)) through vector v = 2 NOTE: Enter your answer in three decimal places. Possible Answers : 0.904 to 0.910

Find the element $P_{2,2}$ of the projection matrix P that project on to the line through vector $v = \begin{bmatrix} -1 \\ 7 \\ 2 \end{bmatrix}$. (Assume matrix entries are indexed starting from (1,1))

MLF-Q027 | [SA] | Marks: +3 | *Appeared in 1 papers*

Find the element Pl.2 of the projection matrix P that project on to the line 2 1 . (Assume matrix entries are indexed starting from (1, 1)). through vector v = —6 NOTE: Enter your answer in three decimal places. Possible Answers : 0.045 to 0.052

Find the element $P_{1,2}$ of the projection matrix P that project on to the line through vector $v = \begin{bmatrix} 2 \\ 1 \\ -6 \end{bmatrix}$. (Assume matrix entries are indexed starting from (1,1)).

MLF-Q057 | [MSQ] | Marks: +4 | *Appeared in 1 papers*

Time : 0 Which of the following statements is/are true?

(A) There exists a n x n matrix A, n is odd, for which the rank of A is not equal to the nullity of A.

There exists a $n \times n$ matrix A, n is odd, for which the rank of A is not equal to the nullity of A.

(B) There exists a n x n matrix A, n is odd, for which the rank of A is equal to the nullity of A.

There exists a $n \times n$ matrix A, n is odd, for which the rank of A is equal to the nullity of A.

(C) There exists a n x n matrix A, n is even, for which the rank of A is not equal to the nullity of A.

There exists a $n \times n$ matrix A, n is even, for which the rank of A is not equal to the nullity of A.

(D) There exists a n x n matrix A, n is even, for which the rank of A is equal to the nullity of A.

There exists a $n \times n$ matrix A, n is even, for which the rank of A is equal to the nullity of A.

Correct Answer: A, C, D

MLF-Q072 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

12 Find a matrix P such that P¯IAP is diagonal. Time : 0

Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$. Find a matrix P such that $P^{-1}AP$ is diagonal.

23

(A) $\begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$

(D) $\begin{bmatrix} -2 & -2 \\ 0 & 0 \end{bmatrix}$

Correct Answer: B

MLF-Q086 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

Which of the following options is/are true when the vector 1 onto the vector 0 1 2 1 projected

$$\text{Which of the following options is/are true when the vector } \begin{bmatrix}1\\2\\1\end{bmatrix} \text{ projected onto the vector } \begin{bmatrix}1\\-1\\0\end{bmatrix}?$$

(A) The projection matrix is —

$$\text{The projection matrix is } \frac{1}{2}\begin{bmatrix}-1 & -1 & 0\\1 & 1 & 0\\0 & 0 & 0\end{bmatrix}$$

(B) The projection matrix is —

$$\text{The projection matrix is } \frac{1}{2}\begin{bmatrix}1 & -1 & 0\\-1 & 1 & 0\\0 & 0 & 0\end{bmatrix}$$

(C) Projected vector is —

$$\text{Projected vector is } \frac{1}{2}\begin{bmatrix}-1\\1\\0\end{bmatrix}$$

(D) Projected vector is —

$$\text{Projected vector is } \frac{1}{2}\begin{bmatrix}1\\-1\\0\end{bmatrix}$$

Correct Answer: B, C

MLF-Q093 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

If B is diagonalizable, what is the invertible matrix P such that B = PDP–1?

Consider a 3×3 matrix B given by

$$B = (-1)\begin{bmatrix} -4 & 3 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & -1 \end{bmatrix}.$$

(A) 400 010 001

$\begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(B) 321 211 101

$\begin{bmatrix} 3 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

(C) 111 210 001

$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(D) B is not diagonalizable.

B is not diagonalizable.

(E) B is not diagonalizable.

B is not diagonalizable.

Correct Answer: D

MLF-Q094 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

If B is diagonalizable, which of the following is a diagonal matrix D such that B = PDP–1? Use the matrix P obtained in the previous question.

Consider a 3×3 matrix B given by

$$B = (-1)\begin{bmatrix} -4 & 3 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & -1 \end{bmatrix}.$$

(A) 200 010 001

$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(B) 400 001

$\begin{bmatrix} 4 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(C) 300 020

$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(D) B is not diagonalizable.

B is not diagonalizable.

(E) B is not diagonalizable.

B is not diagonalizable.

Correct Answer: A

MLF-Q097 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

Consider the following matrices: Which of the following is true?

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

(A) is diagonalizable, is not diagonalizable.

(B) is not diagonalizable, is diagonalizable.

(C) Both and are diagonalizable.

(D) Neither nor is diagonalizable

Correct Answer: B

MLF-Q101 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

Which of the following is/are true?

(A) If and are matrices of the same size, then
$\text{rank}(A+B) = \text{rank}(A) + \text{rank}(B)$

(B) The rank of a matrix is equal to the number of non-zero rows in it.
5×5

(C) If is a matrix which is obtained by permuting the rows of the identity
5×5

(D) If is a non-zero column vector in , then the rank of the matrix is .
$\mathbb{R}^{10}$

Correct Answer: C, D

MLF-Q115 | [MSQ] | Marks: +4 | *Appeared in 1 papers*

Let A, B e are correct? be two matrices, then which among the following statements Time : 0

Let $A, B \in \mathbb{R}^{n \times n}$ be two matrices, then which among the following statements are correct?

(A) x e Nullspace(Ä) e Nullspace(BA).
$x \in \text{Nullspace}(A) \implies x \in \text{Nullspace}(BA).$

(B) e Columnspace(A) a; e Columnspace(B), where A = BC, C e R nxn
$x \in \text{Columnspace}(A) \implies x \in \text{Columnspace}(B)$, where $A = BC, C \in \mathbb{R}^{n \times n}$.

(C) a: e Nullspace(AB) Bx e Nullspace(A).
$x \in \text{Nullspace}(AB) \implies Bx \in \text{Nullspace}(A).$

(D) a; e Nullspace(AB) Nullspace(B).
$x \in \text{Nullspace}(AB) \implies x \in \text{Nullspace}(B).$

Correct Answer: A, B, C

MLF-Q132 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

Consider a matrix A = 010 100 and A = PDP-I. 001 Which of the following options is/are true?

Consider a matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $A = PDP^{-1}$.

Which of the following options is/are true?

(A) A is an orthogonally diagonalizable matrix.

A is an orthogonally diagonalizable matrix.

(B) 1 01

$P = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

(C) —1 0

$P = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$

(D) $P^{-1} = \frac{1}{2}\begin{bmatrix} 1 & 0 & -1 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$

Correct Answer: A, B

MLF-Q144 | [MSQ] | Marks: +4 | *Appeared in 3 papers*

Let P be the projection matrix for a vector a = options is/are true? 1 Which of the following 2

Let P be the projection matrix for a vector $a = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Which of the following options is/are true?

(A) P is orthogonally diagonalizable.

P is orthogonally diagonalizable.

(B) $P^4 = P.$

(C) 1 Projection of b 1 on the vector a is

Projection of $b = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ on the vector a is $-\frac{1}{5}a$.

Rank(P) = 2

(D) $Rank(P) = 2$

Correct Answer: A, B, C

MLF-Q170 | [MCQ] | Marks: +2 | *Appeared in 15 papers*

Which of the following linear relations is valid among the rows (RI , R2, 113) of matrix A?

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Which of the following linear relations is valid among the rows (R_1, R_2, R_3) of matrix A?

(A) $R_2 - R_1 - R_3 = 0$

RI + R2 - 2R3 = O

(B) $R_1 + R_2 - 2R_3 = 0$

O.RI + o.R2 + o.Rg = O

(C) $0.R_1 + 0.R_2 + 0.R_3 = 0$

= 2111 +

(D) $R_2 = 2R_1 + R_3$

Correct Answer: C

MLF-Q173 | [MCQ] | Marks: +2 | *Appeared in 15 papers*

If a square matrix A is diagonalizable, then A is invertible.

(A) True

(B) False

Correct Answer: B

MLF-Q184 | [MCQ] | Marks: +2 | *Appeared in 3 papers*

83 Find a matrix P such that P¯IAP is diagonal. Let = Time : 0

$$\text{Let } A = \begin{bmatrix} 8 & 3 \\ 2 & 7 \end{bmatrix}. \text{ Find a matrix } P \text{ such that } P^{-1}AP \text{ is diagonal.}$$

(A) $\begin{bmatrix} 1 & 1 \\ 2 & -3 \end{bmatrix}$

10 0

(B) $\begin{bmatrix} 10 & 0 \\ 0 & 5 \end{bmatrix}$

11 23

(C) $\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$

(D) $\begin{bmatrix} 3 & 1 \\ 2 & -1 \end{bmatrix}$

Correct Answer: D

■ Category: Span & Basis (2 Questions)

MLF-Q064 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

1 2 Let = 3 5 1 , then which among the following is orthogonal to C(A)? 1 1 Time : 0

$$\text{Let } A = \begin{bmatrix} 1 & 1 \\ 2 & 0 \\ 3 & 1 \\ 5 & 1 \end{bmatrix}, \text{ then which among the following is orthogonal to } C(A)?$$

(A) $\text{Span}\left\{\begin{bmatrix} -3 \\ -5 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}\right\}$

Span 1

(B) $\text{Span}\left\{\begin{bmatrix} 3 \\ 5 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}\right\}$

Span 1

(C) $\text{Span}\left\{\begin{bmatrix} 3 \\ 5 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}\right\}$

Span 1

(D) $\text{Span}\left\{\begin{bmatrix} -3 \\ -5 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}\right\}$

Correct Answer: A

MLF-Q121 | [SA] | Marks: +4 | *Appeared in 1 papers*

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Let P1 be the projection matrix onto the line spanned by the vector a = [1, 2, 3]T . Let P2 be the projection matrix that projects vectors onto the plane perpendicular to span of a. Find the determinant of P1 + P2. Possible Answers : 1 Question Shuffling Allowed : Yes Is Section Default? : null

Category: Subspaces (12 Questions)

MLF-Q063 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

If null space of a matrix A is span can be row space of matrix A? 7/17 22/17 —1 then which of the following Time : 0

If null space of a matrix A is span$\left\{\begin{bmatrix}7/17\\22/17\\-1\end{bmatrix}\right\}$, then which of the following can be row space of matrix A?

Span 17 22 17

(A) $\text{Span}\left\{\begin{bmatrix}1\\0\\\frac{7}{17}\end{bmatrix},\begin{bmatrix}0\\1\\\frac{22}{17}\end{bmatrix}\right\}$

Span , 22 7 17 17

(B) $\text{Span}\left\{\begin{bmatrix}-1\\0\\\frac{7}{17}\end{bmatrix},\begin{bmatrix}0\\1\\\frac{22}{17}\end{bmatrix}\right\}$

Span 17 22 17

(C) $\text{Span}\left\{\begin{bmatrix}1\\0\\\frac{7}{17}\end{bmatrix},\begin{bmatrix}0\\-1\\\frac{22}{17}\end{bmatrix}\right\}$

Span 17 22 17

(D) $\text{Span}\left\{\begin{bmatrix}-1\\0\\\frac{7}{17}\end{bmatrix},\begin{bmatrix}0\\-1\\\frac{22}{17}\end{bmatrix}\right\}$

Correct Answer: A

MLF-Q070 | [MCQ] | Marks: +4 | *Appeared in 3 papers*

Time : 0 Find the row space of matrix A.

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

(A) span $\text{span}\left\{\begin{pmatrix}1\\-1\\2\end{pmatrix}, \begin{pmatrix}2\\-2\\5\end{pmatrix}\right\}$

(B) span $\text{span}\left\{\begin{pmatrix}1\\0\\3\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}\right\}$

(C) span 2 $\text{span}\left\{\begin{pmatrix}1\\2\\3\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}\right\}$

(D) span 2 $\text{span}\left\{\begin{pmatrix}1\\2\\3\end{pmatrix}, \begin{pmatrix}-1\\-2\\-3\end{pmatrix}, \begin{pmatrix}2\\5\\6\end{pmatrix}\right\}$

Correct Answer: B

MLF-Q071 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

Time : 0 Find the column space of matrix A.

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

(A) span $\text{span}\left\{\begin{pmatrix}1\\-1\\2\end{pmatrix}, \begin{pmatrix}2\\-2\\5\end{pmatrix}\right\}$

(B) span $\text{span}\left\{\begin{pmatrix}1\\0\\3\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}\right\}$

(C) span $\text{span}\left\{\begin{pmatrix}1\\-1\\2\end{pmatrix}, \begin{pmatrix}2\\-2\\5\end{pmatrix}, \begin{pmatrix}3\\-3\\6\end{pmatrix}\right\}$

(D) span 2 $\text{span}\left\{\begin{pmatrix}1\\2\\3\end{pmatrix}, \begin{pmatrix}-1\\-2\\-3\end{pmatrix}, \begin{pmatrix}2\\5\\6\end{pmatrix}\right\}$

Correct Answer: A

MLF-Q085 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

Which of the following options is/are true?

(A) O is orthogonal to the vectors 0 and 1

$\begin{bmatrix}0\\0\\1\end{bmatrix}$ is orthogonal to the vectors $\begin{bmatrix}1\\0\\0\end{bmatrix}$ and $\begin{bmatrix}1\\1\\0\end{bmatrix}$.

(B) For a matrix A, vectors of column space of A are perpendicular to the vectors of null space of A.

For a matrix A, vectors of column space of A are perpendicular to the vectors of null space of A.

(C) Length of the vector, a = is 3.

Length of the vector, $a = \begin{bmatrix}1\\1\\-1\end{bmatrix}$ is 3.

(D) 2 and 1 are orthogonal vectors.

$\begin{bmatrix}1\\2\\3\end{bmatrix}$ and $\begin{bmatrix}1\\1\\-1\end{bmatrix}$ are orthogonal vectors.

Correct Answer: A, D

MLF-Q088 | [MSQ] | Marks: +2 | *Appeared in 3 papers*

Consider a matrix 6 15 Which of the following options is/are true?

Consider a matrix

$$A = \begin{bmatrix}2 & 5\\6 & 15\end{bmatrix}.$$

Which of the following options is/are true?

(A) Null space of A is span 2

Null space of A is span $\left\{\begin{bmatrix}-5\\2\end{bmatrix}\right\}$

(B) Column space of A is span 1 3

Column space of A is span $\left\{\begin{bmatrix}1\\3\end{bmatrix}\right\}$

(C) -3 Left null space of A is span 1

Left null space of A is span $\left\{\begin{bmatrix}-3\\1\end{bmatrix}\right\}$

(D) 2 Row space of A is span

Row space of A is span $\left\{\begin{bmatrix}2\\-5\end{bmatrix}\right\}$

Correct Answer: A, B, C

MLF-Q099 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

Which of the following is a basis for the column space of the matrix given below?

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

(A) {(1,2, 1), (1, —1, 0)}
$\{(1,2,1),(1,-1,0)\}$

(B) {(1,0, 0), (0, 1, 0)}
$\{(1,0,0),(0,1,0)\}$

(C) {(1,2, 1), (1, —1, 0), (2, 1, 1)}
$\{(1,2,1),(1,-1,0),(2,1,1)\}$

(D) {(1,3,2), (1, —1, 0)}
$\{(1,3,2),(1,-1,0)\}$

Correct Answer: A

MLF-Q112 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

Time : 0 Which of the following statements is/are true?

(A) Row space of a matrix is orthogonal to the null space of the matrix.

(B) Row space of a matrix is orthogonal to the null space of transpose of the

(C) Dimension of row space of a matrix is always equal to the dimension of

(D) Dimension of row space of a matrix need not be equal to the dimension of

Correct Answer: A, B, C

MLF-Q126 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

Consider a matrix A — 111 021. 132 Which of the following options is/are true?

$$\text{Consider a matrix } A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 1 & 3 & 2 \end{bmatrix}. \text{ Which of the following options is/are true?}$$

(A) Dimension of the column space of A is 1

$$\text{Dimension of the column space of } A \text{ is } 1$$

(B) Dimension of the column space of A is 1

$$\text{Dimension of the column space of } A \text{ is } 1$$

(C) Left null space of A is span —1 —1 2

$$\text{Left null space of } A \text{ is span } \left\{ \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} \right\}$$

(D) Row space of A is span 1 0.5

$$\text{Row space of } A \text{ is span } \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0.5 \end{bmatrix} \right\}$$

Correct Answer: A, C, D

MLF-Q129 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

Consider a subspace S = y, z) I x + 2y -+- 2z = 0} of the vector space kg. Which of the following options is/are true?

$$\text{Consider a subspace } S = \{(x, y, z) \mid x + 2y + 2z = 0\} \text{ of the vector space } \mathbb{R}^3.$$
$$\text{Which of the following options is/are true?}$$

(A) 0 1 is a vector of S. 1

$$\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \text{ is a vector of } S.$$

(B) The dimension of S is 1.

$$\text{The dimension of } S \text{ is } 1.$$

(C) The dimension of S is 1.

$$\text{The dimension of } S \text{ is } 1.$$

(D) is a basis of S. -2.5

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ -2.5 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\} \text{ is a basis of } S.$$

Correct Answer: B, D

MLF-Q169 | [MCQ] | Marks: +3 | *Appeared in 15 papers*

Which of the following sets forms a basis for the row space of A?

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

(A) {(1, 0, 2), (0, 1,–1)}

(B) {(1, 0, 2), (0, 1,–1), (0, 0,–1)}

(C) {(1, 0, 2), (0,–1, 1), (0,–1,–4)}

(D) {(0, 1,–1), (0, 0,–4)}

Correct Answer: B, C

MLF-Q192 | [MSQ] | Marks: +4 | *Appeared in 3 papers*

Time : 0 Select the correct options from the following.

(A) There exists a matrix whose row space contains O and whose null space contains I

There exists a matrix whose row space contains $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ and whose null space contains $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.

(B) There exists a matrix with row space = and column space = IR4.

There exists a matrix with row space = $\mathbb{R}^3$ and column space = $\mathbb{R}^4$.

(C) There exists a matrix whose column space contains O O and row space contains

There exists a matrix whose column space contains $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ and row space contains $\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

(D) There exists a matrix whose nullspace consists of all vectors in RS such that + '2x2 + 4:rg = O.

There exists a matrix whose nullspace consists of all vectors in $\mathbb{R}^3$ such that $x_1 + 2x_2 + 4x_3 = 0$.

Correct Answer: C, D

MLF-Q198 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

Which of the following 3 × 3 matrices have rank 2 with column space equal to row space?

(A) 100 010 ooo

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

(B) 123 245 356

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

(C) ooo ooo ooo

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

(D) 200 ooo 002

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Correct Answer: A, D

■ Category: Vectors (1 Questions)

MLF-Q183 | [MCQ] | Marks: +2 | *Appeared in 3 papers*

Consider a vector v = 1 Does v belong to the columnspace of A — Time : 0

$$\text{Consider a vector } v = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}. \text{ Does } v \text{ belong to the columnspace of } A = \begin{bmatrix} -8 & -2 & -9 \\ 6 & 4 & 8 \\ 4 & 0 & 4 \end{bmatrix}?$$

(A) Yes

(B) No

Correct Answer: A

■ WEEK 4 - REGRESSION & EIGENVALUES

■ Category: Eigenvalues (20 Questions)

MLF-Q025 | [SA] | Marks: +5 | *Appeared in 1 papers*

31 The two eigenvalues of the matrix have a ratio 2:1 for p=3. What is another value of p for which eigenvalues have same ratio 2:1? NOTE: Enter your answer in one decimal place. Possible Answers : 4.4 to 4.6

$$\text{The two eigenvalues of the matrix } \begin{bmatrix} 3 & 1 \\ 1 & p \end{bmatrix} \text{ have a ratio 2:1 for p=3. What is another value of p for which eigenvalues have same ratio 2:1?}$$

MLF-Q058 | [MSQ] | Marks: +4 | *Appeared in 1 papers*

Let P be the matrix that projects vectors in onto the subspace spanned by the vector a = [1, 2, .31. Which of the following are eigenvectors of P corresponding to the eigenvalue 0? Time : 0

$$\text{Let } P \text{ be the matrix that projects vectors in } \mathbb{R}^3 \text{ onto the subspace spanned by the vector } a = [1, 2, 3]^T. \text{ Which of the following are eigenvectors of } P \text{ corresponding to the eigenvalue } 0?$$

(A) [1,–2, 2]

(B) [2, 2,–1]

(C) [–2, 1, 0]

(D) [–3, 0, 1]

Correct Answer: C, D

MLF-Q062 | [MCQ] | Marks: +4 | *Appeared in 1 papers*

Time : 0 The characteristic polynomial of a matrix with eigenvalues 1, 2, 3 and 4 is

(A) - ION + - + 24

$$\lambda^4 - 10\lambda^3 + 35\lambda^2 - 50\lambda + 24$$

(B) + + + + 24

$$\lambda^4 + 10\lambda^3 + 35\lambda^2 + 50\lambda + 24$$

(C) + ION - + - 24

$$\lambda^4 + 10\lambda^3 - 35\lambda^2 + 50\lambda - 24$$

(D) 2M - + - loon + 48

$$2\lambda^4 - 20\lambda^3 + 35\lambda^2 - 100\lambda + 48$$

Correct Answer: A

MLF-Q077 | [MSQ] | Marks: +4 | *Appeared in 3 papers*

Consider a matrix B = A 4 — 3A — 21, where A = options is/are true? 111 0 2 1 . Which of the following 003 Time : 0

$$\text{Consider a matrix } B = A^4 - 3A - 2I, \text{ where } A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}. \text{ Which of the following options is/are true?}$$

(A) Eigenvalues of A are 1, 2, and 3.

(B) Eigenvalues of A are 1, 2, and 2.

(C) Eigenvalues of B are 2, 20, and 78.

(D) Matrix B is diagonalizable.

Correct Answer: A, D

MLF-Q078 | [MSQ] | Marks: +4 | *Appeared in 3 papers*

Time : 0 Let A and B be symmetric matrices such that AB = BA. Then which of the following options is/are true?

(A) AB is a symmetric matrix.

(B) AB is orthogonally diagonalizable.

(C) BA has real eigenvalues.

(D) AB is not a symmetric matrix.

Correct Answer: A, B, C

MLF-Q082 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

Consider the matrix M given by: 12 Which of the following options is true?

Consider the matrix M given by:

$$M = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$$

Which of the following options is true?

(A) Eigenvectors corresponding to the two different eigenvalues of M are orthogonal.

(B) The matrix M is not diagonalizable.

(C) The eigenvalues of M are 4 and 1

(D) The rank of M is 1.

Correct Answer: D

MLF-Q100 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

The matrix has eigenvalues and . Which of the following options is/are correct?

$$P = \begin{bmatrix} m & 2 \\ 1 & n \end{bmatrix}$$

(A) mn = O

$$mn = 0$$

(B) mn = 2

$$mn = 2$$

(C) rn + n = 3

$$m + n = 3$$

(D) —2

$$m + n = 2$$

Correct Answer: B, C

MLF-Q103 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

Which of the following is/are true ?

(A) Every square matrix with real entries has at least one real eigenvalue.

(B) A set of two eigenvectors corresponding to the same eigenvalue of a square matrix must be linearly dependent.

(C) Given a square matrix of order , the set of all eigenvectors for an eigenvalue $\mathbb{R}^n$

(D) A square matrix is not invertible if and only if is one of its eigenvalues.

Correct Answer: C, D

MLF-Q109 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

If is the smallest eigenvalue of , enter the value of . Your answer should be an integer. Possible Answers : -3

$$A = \begin{bmatrix} 1 & 5 \\ 4 & 2 \end{bmatrix}$$

MLF-Q114 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

Let the matrix A e Rdxd be diagonalizable, then which of the following statements is/are true for A? Time : 0

Let the matrix $A \in \mathbb{R}^{d \times d}$ be diagonalizable, then which of the following statements is/are true for A?

(A) A must be symmetric.

A must be symmetric.

(B) We can find a basis of eigenvectors for Rd

We can find a basis of eigenvectors for $\mathbb{R}^d$.

(C) We can find an orthonormal basis of eigenvectors for Rd

We can find an orthonormal basis of eigenvectors for $\mathbb{R}^d$.

(D) A can have some repeated eigenvalues but must have distinct independent eigenvectors.

A can have some repeated eigenvalues but must have distinct independent eigenvectors.

Correct Answer: B, D

MLF-Q128 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

Consider a matrix A — 121 101 001 Which of the following options is/are true?

$$\text{Consider a matrix } A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}. \text{ Which of the following options is/are true?}$$

(A) 1 is an eigenvalue of A.

1 is an eigenvalue of A.

(B) 2 is an eigenvalue of A.

2 is an eigenvalue of A.

(C) -3 —1 is an eigenvector of A. 2

$\begin{bmatrix} -3 \\ -1 \\ 2 \end{bmatrix}$ is an eigenvector of A.

(D) 1 1 is an eigenvector of A. 0

$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ is an eigenvector of A.

Correct Answer: A, B, C

MLF-Q135 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

Which of the following options is/are true ?

$$\text{Consider a matrix } A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}. \text{ Based on this information, answer the given subquestions.}$$

(A) 1 is an eigenvalue of A.

1 is an eigenvalue of A.

(B) 1 + ivo is an eigenvalue of A.

$1 + i\sqrt{2}$ is an eigenvalue of A.

(C) The rank of the matrix A is 2.

The rank of the matrix A is 2.

(D) Nullity of the matrix A is 1.

Nullity of the matrix A is 1.

Correct Answer: A, B

MLF-Q142 | [MSQ] | Marks: +4 | *Appeared in 3 papers*

Which of the following options is/are true?

$$\text{Let } A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \text{ be a } 3 \times 3 \text{ matrix. Use this information to answer the given subquestions}$$

(A) A is a symmetric matrix.

(B) A is a diagonal matrix.

(C) A is orthogonally diagonalizable.

(D) Eigenvalues of A are -1, -1 and 2.

Correct Answer: A, B, C, D

MLF-Q153 | [MCQ] | Marks: +4 | *Appeared in 1 papers*

MSQ Correct Marks : 4 Max. Selectable Options : 0 Which of the following options is/are true?

(A) A is a symmetric matrix.

(B) A is a diagonal matrix.

(C) A is orthogonally diagonalizable.

(D) Eigenvalues of A are -1, -1 and 2.

MLF-Q172 | [MCQ] | Marks: +3 | *Appeared in 15 papers*

Which of the following is true?

$$\text{Consider the matrix } A \text{ given below:}$$

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$\text{where } 0 < \theta < 180^\circ.$$

(A) A does not have real eigenvalues

(B) A has two distinct real eigenvalues

(C) A has two real eigenvalues, not necessarily distinct

Correct Answer: A, B

MLF-Q174 | [MCQ] | Marks: +1 | *Appeared in 15 papers*

The eigenvalues of a real symmetric matrix are real.

(A) True

(B) False

Correct Answer: A, B

MLF-Q175 | [MCQ] | Marks: +2 | *Appeared in 15 papers*

A square matrix A of shape n x n is diagonalizable if and only if it has n distinct eigenvalues.

(A) True

(B) False

Correct Answer: B

MLF-Q191 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

Let A be a n x n real matrix. Then, which among the following statements are true? Time : 0

Let A be a $n \times n$ real matrix. Then, which among the following statements are true?

(A) If the eigenvalues of A are /\1, ... , An, then A is similar to a diagonal matrix with A1, . , along the diagonal.

If the eigenvalues of A are $\lambda_1, \ldots, \lambda_n$, then A is similar to a diagonal matrix with $\lambda_1, \ldots, \lambda_n$ along the diagonal.

(B) If rank(A) = r, then A has r non-zero eigenvalues.

If $\text{rank}(A) = r$, then A has r non-zero eigenvalues.

(C) A may have complex eigenvalues.

A may have complex eigenvalues.

(D) If A has a repeated eigenvalue, then A is not diagonalizable.

If A has a repeated eigenvalue, then A is not diagonalizable.

Correct Answer: C

MLF-Q199 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

Consider a matrix B = A 4 — .3A — 21, where A = following options is/are true? 111 021 003 Which of the

Consider a matrix $B = A^4 - 3A - 2I$, where $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$. Which of the following options is/are true?

(A) B has eigenvalues -4, 8, 70.

(B) A has eigenvalues 1, 16, 81.

(C) B is an upper triangular matrix.

(D) B is a lower triangular matrix.

Correct Answer: A, B, C, D

MLF-Q204 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

Let = 200 021 006 Is A diagonalizable?

$$\text{Let } A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 6 \end{bmatrix}. \text{ Is } A \text{ diagonalizable?}$$

(A) Yes

(B) No

(C) Insufficient information

Correct Answer: A, B

■ Category: Eigenvectors (6 Questions)

MLF-Q110 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

If is an eigenvector of corresponding to , find . Enter your answer correct to two decimal places. Possible Answers : 1.24 to 1.26

$$A = \begin{bmatrix} 1 & 5 \\ 4 & 2 \end{bmatrix}$$

$$v = (a, -1)$$

MLF-Q113 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

Which of the following matrices have eigenvectors as 2 1 2 1 —2 and 1 —1 Time : 0

Which of the following matrices have eigenvectors as $\begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$?

(A) 324 202 $\begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$

(B) 648 846 $\begin{bmatrix} 6 & 4 & 8 \\ 4 & 0 & 4 \\ 8 & 4 & 6 \end{bmatrix}$

(C) 214 846 $\begin{bmatrix} 2 & 1 & 4 \\ 4 & 0 & 4 \\ 8 & 4 & 6 \end{bmatrix}$

(D) 310 202 423 $\begin{bmatrix} 3 & 1 & 0 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$

Correct Answer: A, B

MLF-Q136 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

Which of the following options is/are true ?

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

(A) Nlatrix A is diagonalizable.

Matrix A is diagonalizable.

(B) Eigenvectors of A are orthogonal.

Eigenvectors of A are orthogonal.

(C) If A is diagonalizable and diagonal form of A = P¯IDP, then P is an orthogonal matrix.

If A is diagonalizable and diagonal form of $A = P^{-1}DP$, then P is an orthogonal matrix.

(D) [Option (D)]

Correct Answer: A, B, C

MLF-Q141 | [MSQ] | Marks: +4 | *Appeared in 3 papers*

Which of the following options is/are true ?

Let $A = \begin{pmatrix} 2 & 1 & 1 \\ a & 3 & 2 \\ 3 & b & c \end{pmatrix}$ be a 3×3 matrix. Let α be the eigenvalue corresponding to the eigenvector $v_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ and β is the eigenvalue corresponding to the eigenvector $v_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ of matrix A. Based on this information, answer the given sub-questions.

(A) Eigenvectors, VI and v2 are linearly independent.

Eigenvectors, v_1 and v_2 are linearly independent.

(B) Eigenvectors, VI and v2 are orthogonal.

Eigenvectors, v_1 and v_2 are orthogonal.

(C) 1 A third eigenvector of A is 2 3

A third eigenvector of A is $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

(D) $a + b + c = 9.$

Correct Answer: A, C, D

MLF-Q143 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

Let VI and ve are the eigenvectors of A corresponding to distinct eigen-values. We construct a new matrix B using these eigenvectors as its columns, such that B = [VI, U2]. Based on this information, which of the following options are correct?

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Let v_1 and v_2 are the eigenvectors of A corresponding to distinct eigen-values. We construct a new matrix B using these eigenvectors as its columns, such that $B = [v_1, v_2]$. Based on this information, which of the following options are correct?

(A) Null space of the matrix B is zero space.
Null space of the matrix B is zero space.

(B) Column space of the matrix B is kg.
Column space of the matrix B is $\mathbb{R}^3$.

(C) [Option (C)]

Null space of the matrix B represents a line in IR3.

(D) Null space of the matrix B represents a line in $\mathbb{R}^3$.

Correct Answer: A, C

MLF-Q200 | [MSQ] | Marks: +4 | *Appeared in 3 papers*

Consider a matrix A and a set S Which of the following options is/ are correct?

Consider a matrix $A = \begin{bmatrix} 6 & 3 \\ -2 & 1 \end{bmatrix}$ and a set $S = \{v_1 = (1, -1), v_2 = (1, 1)\}$. Which of the following options is/are correct?

(A) S is the set of eigenvectors of A.
S is the set of eigenvectors of A.

(B) Only VI is an eigenvector of A.
Only v_1 is an eigenvector of A.

(C) 350 0 50 1
$A^{50} = \begin{pmatrix} 1 & 3 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} 3^{50} & 0 \\ 0 & 4^{50} \end{pmatrix} \begin{pmatrix} -2 & -3 \\ 1 & 1 \end{pmatrix}$

(D) 350 0
$A^{50} = \begin{pmatrix} 3 & 1 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} 3^{50} & 0 \\ 0 & 4^{50} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -2 & -3 \end{pmatrix}$

Correct Answer: B, C

■ Category: Error Calculations (10 Questions)

MLF-Q002 | [SA] | Marks: +4 | *Appeared in 1 papers*

For the data sets (d, Yi) = = 1 to 5, Consider the regression model f(x) = a: + 1. What is the mean squared loss of f NOTE: Enter your answer in two decimal places. Possible Answers : 0.55 to 0.65

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,4),(5,7)], i = 1$ to 5, Consider the regression model $f(x) = x + 1$. What is the mean squared loss of $f(x)$

MLF-Q023 | [SA] | Marks: +4 | *Appeared in 1 papers*

For the data sets (x , y) = • = 1 to 5, let the regression function be x -+- 1. Consider the regression model f (x) = -+- 1. What is the mean squared loss of f(:r) NOTE: Enter your answer to the nearest integer. Possible Answers : 1

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,3),(5,6)], i = 1$ to 5, let the regression function be $x + 1$. Consider the regression model $f(x) = x + 1$. What is the mean squared loss of $f(x)$

MLF-Q045 | [SA] | Marks: +4 | *Appeared in 1 papers*

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Give the reconstruction error for this encoder decoder pair. The reconstruction error is the mean of the squared distance between the reconstructed input and input. NOTE: Enter your answer to the nearest integer. Possible Answers : 24 to 25

$f(x_1, x_2, x_3) = \frac{x_1+x_2+x_3}{3}$ is used as encoder function and $g(u) = [u, 2u, 3u]$ is used as decoder function for dimensionality reduction of following data set.

X
[1,2,3]
[2,3,4]

MLF-Q098 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

Consider a dataset for a regression problem where each element is of the form , for . is the feature and is the label of the data-point. Let be two regression models such that: Which of the following options is correct?

$$(x_i, y_i)$$

$$i = 1, 2, 3$$

has the lower mean squared loss.

(A) $f_1(x)$

has the lower mean squared loss.

(B) $f_2(x)$

(C) Both models have same mean squared loss.

Correct Answer: B

MLF-Q111 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

Time : 0 Which of the following statements are correct?

(A) = 0 and e R) = 1.

$\mathbf{I}(444\%3 = 0 \text{ and } \pi \in \mathbb{R}) = 1.$

(B) Mean Squared Error (MSE) is a suitable loss function for classification problems.

Mean Squared Error (MSE) is a suitable loss function for classification problems.

(C) The loss obtained after applying encoder and decoder functions to a given data can be negative.

The loss obtained after applying encoder and decoder functions to a given data can be negative.

(D) Gaussian mixture model is a family of distributions for modeling multimodal data.

Gaussian mixture model is a family of distributions for modeling multimodal data.

Correct Answer: A, D

MLF-Q131 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

The table below shows the original values and predicted values obtained by some machine learning algorithm for multiclass classification problems. Compute the squared error loss and 0-1 loss. Which of the following options regarding the loss Function is/are correct? Original values Predicted values —1 o 2 2 3 1 I 0 0 0 3 3

The table below shows the original values and predicted values obtained by some machine learning algorithm for multiclass classification problems. Compute the squared error loss and 0-1 loss. Which of the following options regarding the loss Function is/are correct?

Original values	−1	0	1	2	3	1	0	3
Predicted values	1	1	1	2	1	0	0	3

(A) Square error loss ≤ 0-1 loss

(B) Square error loss ≥ 0-1 loss

(C) The 0-1 loss is 0.25

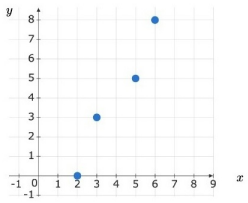
(D) The square error loss is 1.25

Correct Answer: A, B, D

MLF-Q167 | [MCQ] | Marks: +3 | *Appeared in 15 papers*

Which of these models has the smallest squared loss on this training dataset?

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



f@) '2:r — 4

(A) $f(x) = 2x - 4$

(B) $f(x) = 4$

(C) $f(x) = 8$

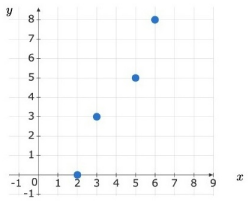
(D) $f(x) = 4 - 2x$

Correct Answer: A

MLF-Q168 | [MCQ] | Marks: +2 | *Appeared in 15 papers*

Is there a model of the form f@) = tvx + b, with w, b IR such that the squared loss is exactly zero on this training dataset?

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Is there a model of the form $f(x) = wx + b$, with $w, b \in \mathbb{R}$ such that the squared loss is exactly zero on this training dataset?

(A) No

(B) Yes

(C) Insufficient information to answer the question

Correct Answer: A, B

MLF-Q182 | [MCQ] | Marks: +2 | *Appeared in 3 papers*

Time : 0 Consider the following input data points: Which among the following models has the least misclassification error?

X	Y
[0, 1]	+1
[0, 2]	+1
[1, 3]	+1
[3, 4]	-1
[4, 5]	-1
[7, 6]	-1

(A) Sign(x1 – 2)

(B) Sign(x2 + 2)

(C) Sign(x1 + x2)

(D) Sign(x2 – x1)

Correct Answer: D

MLF-Q205 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

Consider the following input data points: -1 -1 +1 +1 +1 +1 Which among the following models has the least misclassification error? Hint: Consider Sign(0)

Consider the following input data points:

X	Y
[0, 1, 1]	-1
[0, 1, 0]	-1
[1, 1, 3]	+1
[1, 0, 5]	+1
[0, 0, 6]	+1
[1, 0, 4]	+1

Which among the following models has the least misclassification error?
Hint: Consider $Sign(0) = 1$

(A) $Sign(x_1 + x_2 + x_3 - 5)$

Sign@l + — 2)

(B) $Sign(x_1 + x_2 - 2)$

+ :r.2 + erg)

(C) $Sign(x_1 + x_2 + x_3)$

Sign(rg — $2 — Xl)

(D) $Sign(x_3 - x_2 - x_1)$

Correct Answer: A

■ Category: Least Squares (12 Questions)

MLF-Q051 | [MCQ] | Marks: +4 | *Appeared in 1 papers*

Calculate the projection of vector (2, —1, 3)T onto another vector (—1, 3, I)T Which of the following is the squared length of the projection vector? Time : 0

Calculate the projection of vector $(2, -1, 3)^T$ onto another vector $(-1, 3, 1)^T$.
Which of the following is the squared length of the projection vector?

(A) 0.8686

(B) 0.3636

(C) 1.0363

(D) 1.3636

Correct Answer: B, C

MLF-Q052 | [MCQ] | Marks: +4 | *Appeared in 1 papers*

Time : 0 For the above given data set Raj wants to compute a least square approximation that best fits the dataset with minimum error. Which of the following is the correct equation for the best fit line computed by Raj?

Consider the data set with input x and output y as follows.

x	y
1	2
2	1
1	3
3	4

0 0.412r + 0.215

(A) $\hat{y} = 0.412x + 0.215$

0 = 1.335r + 2.245

(B) $\hat{y} = 1.335x + 2.245$

0.5451* + 1.545

(C) $\hat{y} = 0.545x + 1.545$

1.713r + 1.175

(D) $\hat{y} = 1.713x + 1.175$

0 2.225x + 3.665

(E) $\hat{y} = 2.225x + 3.665$

Correct Answer: C

MLF-Q074 | [MCQ] | Marks: +4 | *Appeared in 3 papers*

Time : 0 Find the best fit line for the dataset given below using least squares method.

x	0	2	3	6	8
y	-2	5	3	1	-3

(A) o.6x 3.2

$$\hat{y} = 0.6x - 3.2$$

(B) = -0.6x + 3.2

$$\hat{y} = -0.6x + 3.2$$

(C) = —x 2.8

$$\hat{y} = -x + 2.8$$

(D) = -0.4x + 2.8

$$\hat{y} = -0.4x + 2.8$$

Correct Answer: B

MLF-Q116 | [MSQ] | Marks: +4 | *Appeared in 1 papers*

Consider the solution of the linear regression problem (Ar — b) using least 155 squares method, where A = 2 3 5 and b = 071 7 8 4 Based on this data, which of the following statements is/are true? Time : 0

Consider the solution of the linear regression problem $(Ax = b)$ using least squares method, where $A = \begin{bmatrix} 1 & 5 & 5 \\ 2 & 3 & 5 \\ 0 & 7 & 1 \end{bmatrix}$ and $b = \begin{bmatrix} 7 \\ 8 \\ 4 \end{bmatrix}$.
Based on this data, which of the following statements is/are true?

(A) The error value is zero as we can get an a: such that Ax is exactly equal to b.
The error value is zero as we can get an x such that Ax is exactly equal to b.

(B) The error value is non-zero as we cannot get an a: such that is exactly equal to b.
The error value is non-zero as we cannot get an x such that Ax is exactly equal to b.

(C) 2 Least squares solution for a; is 0.5 0.5

Least squares solution for x is $\begin{bmatrix} 2 \\ 0.5 \\ 0.5 \end{bmatrix}$.

(D) 1 Least squares solution for a: is 0.6 0.4

Least squares solution for x is $\begin{bmatrix} 1 \\ 0.6 \\ 0.4 \end{bmatrix}$.

Correct Answer: A, C

MLF-Q125 | [SA] | Marks: +3 | *Appeared in 1 papers*

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Using the least square approximation, find the length of the projection vector p of b onto the column space of A. Enter the answer correct to two decimal places. Possible Answers : 3.12 to 3.16 Java

$$\text{Let } A = \begin{pmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{pmatrix} \text{ and } b = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}.$$

MLF-Q133 | [MSQ] | Marks: +4 | *Appeared in 3 papers*

Let j be the regression line (the best fit line) for the dataset given below using the least squares method. Which of the following options is/are true?

Let $\hat{y}$ be the regression line (the best fit line) for the dataset given below using the least squares method. Which of the following options is/are true?

x	0	2	3
y	-2	5	4

(A) 1 = —(13x - 19) 14

$$\hat{y} = \frac{1}{14}(13x - 19)$$

(B) 1 = —(31x - 91) 14

$$\hat{y} = \frac{1}{14}(31x - 91)$$

(C) If = 11, then predicted value of y is '23.

If $x = 11$, then predicted value of y is 23.

(D) 99 If :r = 7, then predicted value of y is — 7

If $x = 7$, then predicted value of y is $\frac{99}{7}$.

Correct Answer: C, D

MLF-Q145 | [MCQ] | Marks: +4 | *Appeared in 3 papers*

Find the best-fit line for the dataset given below using the least squares method. x 0235 •uuuu

Find the best-fit line for the dataset given below using the least squares method.

x	0	2	3	5
y	1	2	4	3

j = 1.34 +

(A) $\hat{y} = 1.34 + 0.46x$

j = 0.46 + 1.34x

(B) $\hat{y} = 0.46 + 1.34x$

= 1.34 - 0.46x

(C) $\hat{y} = 1.34 - 0.46x$

j = 2.16 + 0.74x

(D) $\hat{y} = 2.16 + 0.74x$

Correct Answer: A

MLF-Q162 | [MCQ] | Marks: +3 | *Appeared in 15 papers*

Consider the dataset (r, y) : (—1, 2), (0, 1), (1, 2). Perform polynomial regression of degree 2 by fitting a quadratic model of the form y 00 + + 02:r2. Using the method of least squares, find the estimates of 00, 01 and 02.

Consider the dataset $(x, y) : (-1, 2), (0, 1), (1, 2)$. Perform polynomial regression of degree 2 by fitting a quadratic model of the form $y = \theta_0 + \theta_1 x + \theta_2 x^2$. Using the method of least squares, find the estimates of θ_0, θ_1 and θ_2.

(A) $y = x^2 + 1$

(B) $y = x^2$

(C) $y = x^2 - 1$

(D) $y = 2x$

Correct Answer: A

MLF-Q190 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

Find the best fit line for the dataset given below using least squares method. 13457 Time : 0

Find the best fit line for the dataset given below using least squares method.

x	1	3	4	5	7
y	-3	7	1	0	4

j = 0.7x — 1

(A) $\hat{y} = 0.7x - 1$

j = x + 0.5

(B) $\hat{y} = x + 0.5$

0.2

(C) $\hat{y} = x + 0.2$

(D) $\hat{y} = 0.2x + 1$

Correct Answer: A

MLF-Q193 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

Time : 0 Find the projection of b onto U.

Consider a subspace $U \subseteq \mathbb{R}^3$ and a vector $b \subseteq \mathbb{R}^3$, where

$$U = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}, \quad b = \begin{bmatrix} 2 \\ 0 \\ -4 \end{bmatrix}$$

(A) $\begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$

(B) $\begin{bmatrix} -1 \\ -1 \\ -2 \end{bmatrix}$

(C) $\begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$

(D) $\begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}$

Correct Answer: B

MLF-Q194 | [SA] | Marks: +1 | *Appeared in 3 papers*

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Find the length of the error vector e =b – p, where p is the projection of b onto U. Enter the answer correct to two decimal places. Possible Answers : 3.70 to 3.78 Java

Consider a subspace $U \subseteq \mathbb{R}^3$ and a vector $b \subseteq \mathbb{R}^3$, where

$$U = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}, \quad b = \begin{bmatrix} 2 \\ 0 \\ -4 \end{bmatrix}$$

MLF-Q202 | [MSQ] | Marks: +4 | *Appeared in 3 papers*

Which of the following options is/are correct for the following dataset? auuu nuuu

Which of the following options is/are correct for the following dataset?

x	1	1	-1
y	1	1	1

(A) The best-fit line for the dataset given using the least squares method is, j = 1,
The best-fit line for the dataset given using the least squares method is, $\hat{y} = 1$.

(B) The best-fit line for the dataset given using the least squares method IS, y
The best-fit line for the dataset given using the least squares method is, $\hat{y} = -1$.

(C) The best-fit line for the dataset given using the least squares method is, j = — 1.
The best-fit line for the dataset given using the least squares method is, $\hat{y} = x - 1$.

(D) The best-fit line for the dataset given using the least squares method is,
The best-fit line for the dataset given using the least squares method is, $\hat{y} = x + 1$.

(E) Error vector (the difference between predicted value and actual value) is 0 -2
Error vector (the difference between predicted value and actual value) is $\begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix}$.

(F) 0 Error vector(the difference between predicted value and actual value) is 0 0
Error vector(the difference between predicted value and actual value) is $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Correct Answer: A, F